INDUCED PATHS, RADIUS, AND EXTREMAL NEIGHBORHOOD INDEPENDENCE IN GRAPHS WITHOUT 4-CYCLES

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Abstract. Let $M(n)$ be the maximum of $\ell(G) = S(G)/n$ over connected graphs of order n with no 4-cycle subgraph and no induced path on six vertices, where $S(G)$ is the sum of $\alpha(G[N(v)])$. Then $M(7) = 16/7$, $M(8) = 5/2$, $M(9) = 8/3$, $M(10) = 3$ and $M(n) = 2 + 2/n$ for every $n \geq 11$. The extremal graph is unique at each order. The maximum is the value 3, attained only by the Petersen graph. Deleting one vertex of the Petersen graph gives the unique extremal graph of order 9, and a single graph H_8 is extremal at order 8. For every $n \geq 11$ the unique extremal graph consists of a 6-cycle, one further vertex joined to an antipodal pair, and $n - 7$ leaves at that vertex. In particular $S \leq 3n$, with equality only for the Petersen graph.

A connected graph with no 4-cycle subgraph, of radius 4 and induced-path order 7, is either C_8 or is obtained from a 6-cycle by a matching-and-isolate extension at each cycle vertex, together with at most one additional vertex on each cycle edge. For every such graph $S = 2n - b - m$, and $S = 2n$ precisely for C_8 and for the graphs obtained from C_6 by attaching at least one leaf at every cycle vertex.

Every connected triangle-free graph with no 4-cycle subgraph and no induced path on seven vertices satisfies $2e \leq 3n$, with equality only for the Petersen graph.

These three statements do not use Conjecture 133 of *Written on the Wall II*. As an application, the theorem of Erdős, Saks and Sós yields that conjecture, in the real-valued form $p(G) \geq r(G) + \ell(G)$, whenever $\ell \leq 3$, whenever $r \leq 2$ and $\ell \leq 4$, whenever $r = 4$ and $\ell \leq 4$, and whenever $r \geq 5$ and $\ell \leq 4$. Every remaining counterexample with $\ell \leq 4$ lies in the cell $r = 3$, $p = 6$ and $3 < \ell \leq 4 - 9/n$, and every graph in that cell contains a triangle. The value $\ell = 4$ does not occur for a connected graph with no 4-cycle subgraph and no induced path on seven vertices. The conjecture itself remains open.

1. Introduction

All graphs are finite and simple. For a graph G with $n \geq 2$ vertices let $p(G)$ be the largest number of vertices of an induced path, $r(G)$ the radius, and

$$\ell(G) = \frac{1}{n} \sum_{v \in V(G)} \alpha\big(G[N(v)]\big),$$

where α is the independence number and $N(v)$ the open neighbourhood. Write $S(G)$ for the sum in the numerator, and write $c_{C_4}(G) = 1$ if G contains no 4-cycle as a subgraph and $c_{C_4}(G) = 0$ otherwise. A graph with no 4-cycle subgraph is called *strong C_4-free*.

The two structural results of the paper are Theorems 3.6 and 4.3. Let P denote the Petersen graph and $P - v$ the graph obtained by deleting one vertex. Theorem 3.6 determines the maximum $M(n)$ of ℓ over connected strong C_4-free P_6-free graphs. The value is $16/7$ at order 7, $5/2$ at order 8, attained only by the graph H_8 of Figure 1, $8/3$ at order 9, attained only by $P - v$, and 3 at order 10, attained only by P. For every $n \geq 11$ the maximum is $2 + 2/n$, attained only by the graph in that figure: a 6-cycle, a vertex a joined to one antipodal pair, and $n - 7$ leaves at a. The same theorem gives $S \leq 3n$, with equality only for P. Theorem 4.3 describes every connected strong

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C_4-free graph of radius 4 and induced-path order 7, and Corollaries 4.4 and 4.5 give S. Neither statement is a special case of the conjecture recorded next.

Conjecture 1.1 (Graffiti/WOWII Conjecture 133)**.** *Every connected graph G on at least two vertices satisfies*

$$r(G) + \ell(G)^{\,c_{C_4}(G)} \le p(G).$$

The conjecture is number 133 in E. DeLaVina's list *Written on the Wall II* of conjectures of the program Graffiti.pc [7]. We follow the formal statement in the Formal Conjectures repository at the fixed commit `5faef0c4dc3d226f8f9a9fc8f581b6221283eb6e` [10], the merge commit of pull request #5727 [11]. That commit removed the floor function from an earlier version of the file. The statement there reads `(rad : ℝ) + 1 G ^ cC4 ≤ (path G : ℝ)`. We work with this real-valued statement and call it REAL-133.

If G contains a 4-cycle, the conjecture reads $p(G) \ge r(G) + 1$. A shortest path between two vertices at distance $\operatorname{diam}(G) \ge r(G)$ is induced, so the inequality holds. Only strong C_4-free graphs remain. Since p and r are integers, REAL-133 for such a graph is equivalent to

$$p(G) - r(G) \;\ge\; \lceil \ell(G) \rceil. \tag{1}$$

Theorem 5.9 is the range in which the estimates of this paper give REAL-133. The triangle-free bound is Theorem 5.4. With Lemma 5.6, every counterexample in the range $\ell \le 4$ falls in the cell of Corollary 5.7.

H. Chen [3] studies the floor form $p \ge r + \lfloor \ell \rfloor$ of the same conjecture. The arguments below do not use his proofs. The statements that describe the same configurations are the following. His Lemma 2.1 is the local matching lemma: in a strong C_4-free graph the neighbourhood of every vertex is a disjoint union of edges and isolated vertices, and $S = 2e - 3T$. Lemma 2.1 records the same description. His Lemma 7.18 states that if an induced 6-cycle lies in a strong C_4-free P_6-free graph, then every vertex outside the cycle has either no neighbour on the cycle or exactly two, and those two are antipodal. His Corollary 7.20 records the consequence for the cycle vertices: each has triangle count zero, and its degree is 2 or 3 according as its antipodal slot is empty or occupied. The same restriction on traces is Lemma 3.4 when the cycle dominates, and it is the description of the first distance layer in the proof of Theorem 3.6. Immediately after the proof of his Theorem 3.6, Chen records that every regular graph of girth at least 5 and radius 2 satisfies the floor form, and that this includes every Moore graph of diameter 2. The second assertion of Proposition 8.4 is the corresponding statement for the real form, for every Moore graph of diameter 2 and girth 5.

The exact function $M(n)$ is not a consequence of those lemmas. The proof of Theorem 3.6 continues to the second distance layer, and it treats the induced 5-cycle branch and the chordal branch separately. The value of $M(n)$ and the uniqueness of the extremal graph are obtained at every order. Theorem 5.4 starts from the classification, due to Chen and Xu [5, Lemma 3.4], of the connected triangle-free strong C_4-free P_7-free graphs with no clique cutset. The edge counts of that list, together with the treatment of cut vertices and of clique cutsets of order two, give the sharp bound for every connected graph in the class.

DeLaViña, Larson, Pepper and Waller [8] summarise known criteria for the equality $p = 2r - 1$ and the r-ciliates of Fajtlowicz: the graphs obtained from an even cycle by attaching a pendant path of one common length at every vertex. Examples 9.1 and 9.7 start from that skeleton and add further blocks. In Example 9.1 each added path of length four contributes 3 vertices and 4 edges, so it increases $S = 2e$ by 8 while the radius and the induced-path order remain those of the ciliate. In Example 9.7 each attached copy of the incidence graph of a projective plane adds $(q+1)f$ to S and $f - 1$ to the order, and the radius and the path order remain $t + 3$ and $2t + 5$.

Section 3 proves Theorem 3.6. Section 4 proves Theorem 4.3. Section 5 proves the triangle-free bound; the rooted inequality of order three is placed there because the cut-vertex induction uses

it. Section 6 is the application to REAL-133 and the exclusions in the range $5 < \ell \leq 6$. The remaining rooted inequalities are in Sections 7 and 8. Table 1 is the list of what is proved and what is open. A finite search is not a proof in this article. Where the repository does not contain a versioned checker, an input list, and a completeness argument, the corresponding statement is not used as a hypothesis.

2. Preliminaries

For $v \in V(G)$ let $d(v)$ be its degree and $t(v)$ the number of triangles containing v. Write $\lambda(v) = \lambda_G(v) = \alpha(G[N(v)])$, $S(G) = \sum_v \lambda(v)$, $e(G)$ for the number of edges and $T(G)$ for the number of triangles.

Lemma 2.1. *Let G be strong C_4-free. Then:*

1. *two distinct vertices have at most one common neighbour;*
2. *for every v, $G[N(v)]$ is a matching together with isolated vertices, so $\lambda(v)$ equals the number of components of $G[N(v)]$ and $\lambda(v) = d(v) - t(v)$;*
3. *$S(G) = 2e(G) - 3T(G)$.*

Proof. (1) Two common neighbours w, w' of x, y give the 4-cycle $xwyw'$. (2) A vertex of $N(v)$ with two neighbours in $N(v)$ would have two common neighbours with v. So the components are single vertices and edges, each edge being a triangle at v. (3) Sum $d(v) - t(v)$ over v. □

The components of $G[N(v)]$ are the *units* of v. For a vertex set U, $G-U$ is the subgraph induced by $V \setminus U$, and $D_G(U) = S(G) - S(G - U)$ is the *deletion loss* of U. We write $L(v) = D_G(\{v\})$.

Lemma 2.2. *Let G be strong C_4-free and $\emptyset \neq U \subseteq V(G)$. Then*

$$D_G(U) = \sum_{u \in U} \lambda(u) + \sum_{x \notin U} \#\{\text{units of } x \text{ contained in } U\}.$$

In particular $L(v) = 2\lambda(v) - t(v) = 2d(v) - 3t(v)$.

Proof. For $x \notin U$, deleting U removes exactly the units of x that lie inside U; a unit that is an edge with one end in U survives as a vertex. For $U = \{v\}$: the $2t(v)$ neighbours of v in triangles with v keep their value, and each of the other $d(v) - 2t(v)$ neighbours loses one. The total is $\lambda(v) + d(v) - 2t(v) = 2\lambda(v) - t(v)$. □

Lemma 2.3. *Let v be a cut vertex of G. Split the components of $G - v$ into two nonempty groups, and let G_1, G_2 be the subgraphs induced by the two groups together with v. Then $S(G) = S(G_1) + S(G_2)$.*

Proof. $G[N(v)]$ has no edges between the groups, so $\lambda_G(v) = \lambda_{G_1}(v) + \lambda_{G_2}(v)$. Every other vertex keeps its neighbourhood. □

We use three published results.

Theorem 2.4 (Erdős, Saks and Sós [9])**.** *Every connected graph G has an induced path on at least $2r(G) - 1$ vertices.*

Theorem 2.5 (Camby and Schaudt [1, Theorem 5])**.** *Let $k \geq 4$ and let G be a connected graph with no induced path on k vertices. Then G has a connected dominating set X such that $G[X]$ has no induced path on $k - 2$ vertices or $G[X]$ is isomorphic to the cycle C_k.*

Theorem 2.6 (Seinsche [13])**.** *If a graph on at least two vertices has no induced path on four vertices, then it or its complement is disconnected.*

Theorem 2.5 is the implication (i)⇒(ii) of [1, Theorem 5] applied with $H = G$. We write P_k for the path on k vertices, and "G is P_k-free" means that G has no induced P_k, that is, $p(G) \le k-1$. A connected graph of radius r has at least $2r$ vertices.

Lemma 2.7. *Let H be a connected strong C_4-free P_4-free graph on at least two vertices. Then H has a vertex adjacent to all other vertices.*

Proof. By Theorem 2.6, H is the join of two nonempty vertex sets A and B. If $|A|, |B| \ge 2$, then $a, a' \in A$ and $b, b' \in B$ span the 4-cycle $aba'b'$. So one side is a single vertex, and it is adjacent to everything. □

3. Small-path mass bounds

Lemma 3.1. *If G is connected and strong C_4-free with $r(G) = 1$, then $S(G) = 2(n-1) - m$, where m is the number of edges of $G[N(u)]$ for a dominating vertex u. Hence $\ell(G) < 2$, and (1) holds.*

Proof. Let $v \ne u$. By Lemma 2.1(1), v has at most one neighbour other than u, and such a neighbour lies in $N(u)$. So $G[N(v)]$ is a clique and $\lambda(v) = 1$; also $\lambda(u) = (n-1) - m$. Hence $S = 2(n-1) - m < 2n$ and $\lceil \ell \rceil \le 2$. If G is complete then $G \in \{K_2, K_3\}$, $\ell = 1$ and $p = 2 = r + 1$. Otherwise two nonadjacent vertices of $N(u)$ together with u form an induced P_3, so $p \ge 3 \ge r + \lceil \ell \rceil$. □

Lemma 3.2. *If G is connected and strong C_4-free with $r(G) \ge 2$, then $p(G) \ge 4$.*

Proof. If G were P_4-free, Lemma 2.7 would give a dominating vertex, so $r(G) \le 1$. □

Lemma 3.3. *If G is connected, strong C_4-free and P_5-free, then $S(G) \le 2n - h$, where h is the number of vertices v with $\lambda(v) \ge 3$. In particular $S(G) \le 2n$.*

Proof. Call v *simplicial* if $\lambda(v) = 1$ and *high* if $\lambda(v) \ge 3$. For $x \in N(v)$ put $X_x = N(x) \setminus N[v]$; then $N(x) \subseteq \{v\} \cup \{\bar{x}\} \cup X_x$, where $\bar{x}$ is the partner of x if x lies in an edge unit of v.

Key fact. Let x, y lie in different units of v. A common vertex of X_x and X_y would give the 4-cycle $vxwy$, so $X_x \cap X_y = \emptyset$. If $x' \in X_x$ and $y' \in X_y$ are nonadjacent, then $x'xvyy'$ is an induced P_5: $x \not\sim y$ because they lie in different units, $x \not\sim y'$ and $y \not\sim x'$ by disjointness, and v is adjacent to neither x' nor y'. So X_x is complete to X_y. If both are nonempty and $|X_x| \ge 2$, a vertex of X_y has two neighbours in X_x, which are also neighbours of x, a contradiction to Lemma 2.1(1). Hence if both are nonempty, then $|X_x| = |X_y| = 1$.

Main step. For each high v we find $\lambda(v) - 1$ simplicial vertices whose only high neighbour is v, one in each of $\lambda(v) - 1$ distinct units of v.

Case 1: some $u \in N(v)$ has $|X_u| \ge 2$. By the key fact $X_y = \emptyset$ for every y outside the unit of u. Then $N(y) \subseteq \{v, \bar{y}\}$, which is a clique, so y is simplicial; so is $\bar{y}$. Hence v is the only high neighbour of y, and the $\lambda(v) - 1$ units other than that of u supply the vertices.

Case 2: $|X_x| \le 1$ for all $x \in N(v)$. Then $\lambda(x) \le 2$ for $x \in N(v)$. Call a unit *active* if one of its vertices x has $X_x \ne \emptyset$. Suppose two units are active, with $y' \in X_y$ and $z' \in X_z$, and let x lie in a third unit. By the key fact $y' \sim z'$, and $xvyy'z'$ is an induced P_5: $x \not\sim y$ (different units), $x \not\sim y', z'$ (disjointness), $v \not\sim y', z'$, and $y \not\sim z'$ (disjointness). As $\lambda(v) \ge 3$, a third unit exists, so at most one unit is active. Every vertex of an inactive unit is simplicial and its only high neighbour is v. This gives at least $\lambda(v) - 1$ units.

Counting. The chosen vertices for distinct high vertices are distinct, since each has a unique high neighbour. So $\sum_{v \text{ high}} (\lambda(v) - 1) \le \#\{\text{simplicial vertices}\}$, and

$$S = \sum_v \lambda(v) \le 2n - \#\{\text{simplicial}\} + \sum_{v \text{ high}} (\lambda(v) - 2) \le 2n - h.$$ □

Lemma 3.4. *Let G be connected, strong C_4-free and P_6-free with a dominating induced cycle $C = v_0v_1\cdots v_5$, and let $W = V(G) \setminus V(C)$. Then every $w \in W$ is adjacent to exactly one antipodal pair $\{v_i, v_{i+3}\}$ and to no other vertex of C, distinct vertices of W use distinct pairs, and W is independent. Hence $|W| \le 3$, $S(G) = 12 + 4|W| < 3n$, and $r(G) \le 2$ if $W \neq \emptyset$.*

Proof. Indices are modulo 6. Let $w \in W$; it has a neighbour on C. If w has exactly one neighbour v_i, then $wv_iv_{i+1}v_{i+2}v_{i+3}v_{i+4}$ is an induced P_6. If $N(w) \cap C = \{v_i, v_{i+1}\}$, then $wv_{i+1}v_{i+2}v_{i+3}v_{i+4}v_{i+5}$ is an induced P_6. Two neighbours at distance 2 on C give a 4-cycle. A set of at least three vertices of C_6 contains two vertices at distance 2. So $N(w) \cap C$ is an antipodal pair. Two vertices of W with the same pair give two common neighbours. If $w \sim w'$ with distinct pairs, some $v_i \in N(w)$ and $v_j \in N(w')$ are adjacent on C, and wv_iv_jw' is a 4-cycle. There are no triangles, $e(G) = 6 + 2|W|$, and Lemma 2.1(3) gives $S = 12 + 4|W|$. Finally, if w uses $\{v_i, v_{i+3}\}$, then every vertex of C is within distance 2 of v_i (the vertex v_{i+3} through w), and each $w' \in W$ has a neighbour in $\{v_{i-1}, v_i, v_{i+1}\}$, because each of the three antipodal pairs meets this set. So v_i has eccentricity 2. □

The same trace analysis applies to any vertex adjacent to an induced 6-cycle in a connected strong C_4-free P_6-free graph, whether or not the cycle dominates. Theorem 3.6 uses that form.

Lemma 3.5. *Let G be connected, strong C_4-free and P_6-free. Then either G has a vertex of eccentricity at most 2, or $G = C_6$.*

Proof. Apply Theorem 2.5 with $k = 6$ to get a connected dominating set X. If $G[X] \cong C_6$, Lemma 3.4 gives $G = C_6$ or a vertex of eccentricity 2. Otherwise $G[X]$ is P_4-free. If $|X| = 1$, the vertex in X dominates G. If $|X| \ge 2$, Lemma 2.7 gives $u \in X$ adjacent to all of $X \setminus \{u\}$, and since X dominates G, every vertex is within distance 2 of u. □

Theorem 3.6. *Let $M(n)$ be the maximum of $\ell(G)$ over connected strong C_4-free P_6-free graphs of order $n \ge 7$. Then*

$$M(n) = \begin{cases} 16/7 & \text{if } n = 7, \\ 5/2 & \text{if } n = 8, \\ 8/3 & \text{if } n = 9, \\ 3 & \text{if } n = 10, \\ 2 + 2/n & \text{if } n \ge 11. \end{cases}$$

For $n = 10$ the unique extremal graph is the Petersen graph P. For $n = 9$ it is $P - v$. For $n = 8$ it is the graph H_8 of Figure 1. For $n = 7$ and for every $n \ge 11$, the unique extremal graph is the graph Γ_{n-7} of that figure: a 6-cycle, a vertex a adjacent exactly to one antipodal pair, and $n - 7$ leaves adjacent only to a. In particular, every connected strong C_4-free P_6-free graph G of order $n \ge 2$ satisfies $S(G) \le 3n$, with equality if and only if G is the Petersen graph.

Proof. Throughout, $S = 2e - 3T$ by Lemma 2.1. A chordless cycle of length at least seven contains an induced P_6. Thus the only possible holes have lengths five and six.

Case 1: an induced six-cycle. Label its vertices $c_0, \ldots, c_5$, indices modulo 6, and write A for the set of vertices at distance one from the cycle. The trace argument in the proof of Lemma 3.4 shows that each $a \in A$ meets the cycle in one antipodal pair. Distinct vertices of A use distinct pairs, and A is an independent set: an edge of A joins distinct pairs, and some contact of one end is adjacent on the cycle to some contact of the other, giving a 4-cycle. Hence $k = |A| \le 3$. Figure 1 shows the three extremal graphs of this branch.

Every vertex x at distance two from the cycle is adjacent to every vertex of A. Such an x has at least one neighbour in A. If $x \sim a$ and $x \not\sim b$ for $a, b \in A$, choose contacts so that $c_i \in N(a)$ and $c_{i+2} \in N(b)$. This is possible: the pairs are distinct, so some contact of b lies at cycle distance two

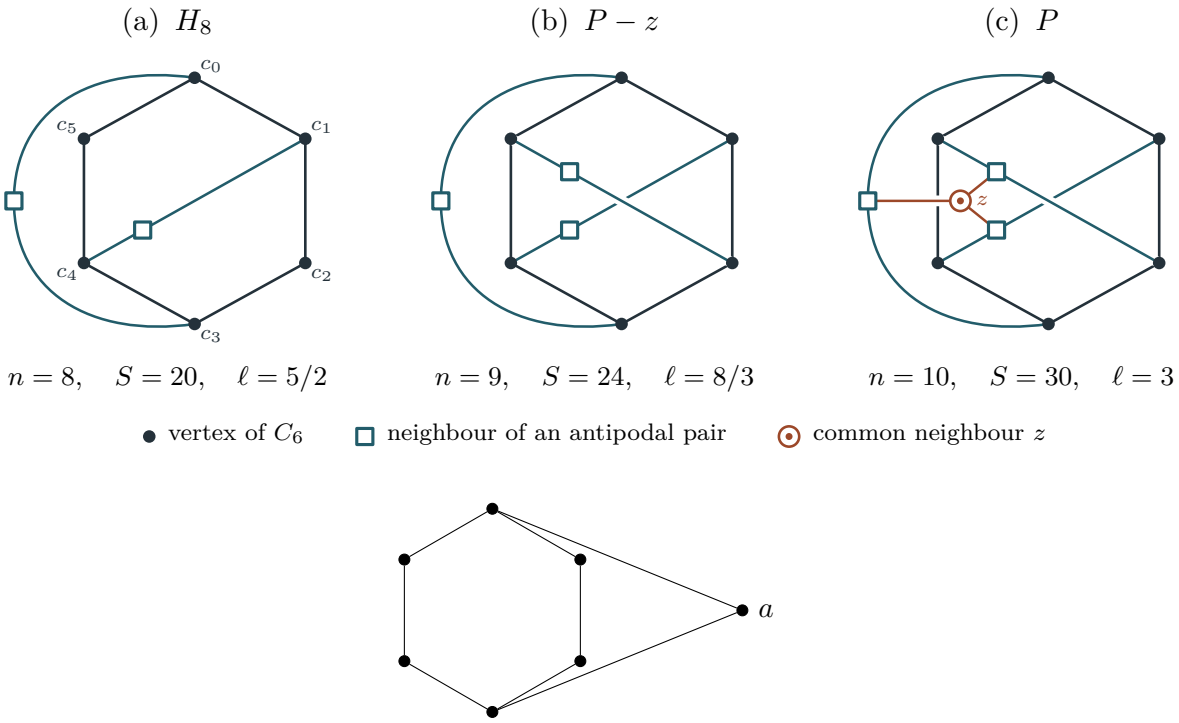


FIGURE 1. Extremal graphs for $M(n)$. In H_8, two antipodal pairs on the 6-cycle have a common neighbour outside the cycle. Adding a common neighbour for the third pair gives $P - z$. Joining a new vertex z to the three outside vertices gives the Petersen graph P. The three graphs are triangle-free, so $S = 2e$. The drawing underneath is Γ_0, the unique graph of order 7 with $\ell = 16/7$: the marked vertex a is adjacent to one antipodal pair. Attaching any number of leaves at a gives the unique extremal graph of each order $n \geq 11$, with $S = 2n + 2$.

from some contact of a, and we orient the cycle from that contact of a toward that contact of b. The path $x, a, c_i, c_{i+1}, c_{i+2}, b$ is then an induced P_6. Indeed x meets the cycle in nothing, a meets the cycle only in its antipodal pair (neither c_{i+1} nor c_{i+2}), b meets the cycle only in its antipodal pair (so $b \sim c_{i+2}$ and $b \not\sim c_i, c_{i+1}$), and $a \not\sim b$.

There is no vertex at distance three or more from the cycle. A shortest path z, x, a, c_i from such a vertex down to the cycle, extended by c_{i+1}, c_{i+2}, is an induced P_6: the vertex z has no neighbour in A or on the cycle, and x has no neighbour on the cycle.

Let h be the number of vertices at distance two. If $k = 0$, then A is empty, so the cycle has no neighbour and $G = C_6$. If $k = 1$, every distance-two vertex is adjacent to the unique vertex $a \in A$. The distance-two vertices induce a matching together with isolated vertices: a vertex of that layer with two neighbours in the layer would have two common neighbours with a. Writing m for the number of edges of the matching gives $n = 7 + h$, $e = 8 + h + m$ and $T = m$, so $S = 16 + 2h - m = 2n + 2 - m$. Thus $S \leq 2n + 2$, with equality if and only if $m = 0$. Equality means that G is the graph Γ_h formed by the 6-cycle, the vertex a on one antipodal pair, and h leaves at a.

The graph Γ_h lies in the class, for every $h \geq 0$. A leaf has degree one. Every other cycle through an outside vertex uses both edges from a to the cycle, and each of the two a-paths along the cycle has length three, so every such cycle has length five. There is no 4-cycle. There is no induced P_6. A leaf, having degree one, occupies an end of any path that contains it, and that path

meets a next. Two leaves therefore determine a path on only three vertices. An induced path that contains one leaf and five further vertices would restrict to an induced path of order five in Γ_0 that starts at a. No such path exists: the two neighbours of a on the cycle are an antipodal pair, the path $a, c_i, c_{i+1}, c_{i+2}, c_{i+3}$ has the chord ac_{i+3}, and every shorter continuation from a through one neighbour stops at the vertex before the antipode, because the next cycle vertex is adjacent to a. An induced path of Γ_0 that avoids a lies on the 6-cycle and has order at most five. An induced path of Γ_0 that contains a in its interior has c_i and c_{i+3} as the two neighbours of a on the path. Each side can then take at most one further cycle vertex: the next vertex on either arc is adjacent to the antipode already used. The resulting path has order at most five. Thus Γ_h is strong C_4-free and P_6-free, and it attains $S = 2n + 2$.

If $k \in \{2, 3\}$, then $h \leq 1$: two vertices of the second layer would both be adjacent to two vertices of A. There are no triangles. The second-layer vertex, if it exists, is adjacent to every vertex of A and to nothing on the cycle, A is independent, and each vertex of A contributes two edges to the cycle. Thus $n = 6 + k + h$ and $S = 2(6 + 2k + kh)$. The four possibilities are

(k, h)	$(2, 0)$	$(2, 1)$	$(3, 0)$	$(3, 1)$
S/n	$5/2$	$8/3$	$8/3$	3

The four pairs are the graphs of Figure 1, with $(3, 0)$ and $(2, 1)$ both equal to $P - z$. Write a_i for the vertex adjacent to the antipodal pair $\{c_i, c_{i+3}\}$. The pair $(k, h) = (3, 1)$ is the graph in panel (c): the three vertices a_0, a_1, a_2 and a vertex z adjacent to all three. It is cubic of order 10. It is triangle-free, and the trace rules give no 4-cycle, so its girth is at least 5. Its diameter is 2. Indeed every pair of cycle vertices is at distance at most 2 along the cycle or through the corresponding a_i; the vertex z meets every a_i; and a non-neighbour of a_i on the cycle is adjacent to c_i or to c_{i+3}. Suppose $v_0 \cdots v_5$ were an induced path. Diameter 2 gives v_0 a common neighbour with each of v_3, v_4, v_5. Those neighbours lie off the path, because the only neighbour of v_0 on the path is v_1 and the path is induced. They are distinct: a vertex adjacent to two of v_3, v_4, v_5 would form a triangle with a path edge, or a 4-cycle with the length-2 subpath $v_3v_4v_5$. This is impossible: v_0 has degree 3, so it has only two neighbours off the path. Thus the graph is P_6-free. It is the Petersen graph P, and $\ell(P) = 3$.

Deleting z leaves $(k, h) = (3, 0)$, the graph $P - z$ in panel (b). The pair $(2, 1)$ is isomorphic to the same graph. In the labelling with $A = \{a_0, a_1\}$ and second-layer vertex z, the cycle $c_0, a_0, c_3, c_4, a_1, c_1$ has its three antipodal pairs served by c_5, z and c_2, and no vertex remains. Both graphs therefore have $\ell = 8/3$, and both are isomorphic to $P - v$. Induced subgraphs of P are strong C_4-free and P_6-free. The pair $(2, 0)$ is the graph H_8 in panel (a), with $\ell = 5/2$. The three choices of the unused antipodal pair differ by a rotation, so this graph is unique up to isomorphism. Its order is 8, so it is neither P nor $P - v$. For $k \in \{2, 3\}$ one has $n = 6 + k + h \leq 10$.

Case 2: an induced five-cycle and no induced six-cycle. Write $C = c_0 \cdots c_4c_0$. Every vertex at distance one from C has a singleton or consecutive-pair trace: any other pair of contacts lies at distance two on C and forms a 4-cycle. If a vertex x were at distance two, with a neighbour a in the first layer, the path x, a followed by four consecutive cycle vertices, beginning at one contact of a and omitting the second contact if a has one, would be an induced P_6. Thus C dominates G.

Write h for the number of vertices with singleton trace and j for the number with a consecutive-pair trace. At most one vertex carries each of the five pair traces, since two would have two common neighbours, so $j \leq 5$. The edges outside C are of the following kinds.

- Two singleton vertices with different supports are nonadjacent. Neighbouring supports give a 4-cycle. Supports at distance two give an induced 6-cycle along the longer arc of C. Edges within one singleton class form a matching, because two neighbours in that class

would be two common neighbours of the supporting cycle vertex. Let m be the number of these edges.

- No two pair vertices are adjacent. From their two distinct support edges one can choose adjacent cycle vertices, one from each edge, and those two vertices together with the pair vertices form a 4-cycle.
- A singleton supported at c_i meets only a pair vertex supported on the opposite edge $\{c_{i+2}, c_{i+3}\}$. Every other pair edge contains a neighbour of c_i and yields a 4-cycle. That pair vertex meets at most one vertex of the singleton class, again by the common-neighbour bound. Let b be the number of singleton-to-pair edges, so $b \le \min(h, j)$.

Consequently $n = 5 + h + j$ and $e = 5 + h + 2j + m + b$. The triangles are the j pair triangles and the m matching triangles, so $T = j + m$ and

$$S = 2e - 3T = 2n + 2b - j - m.$$

The integer b is at most 1. The opposite edge of the support c_i is $\{c_{i+2}, c_{i+3}\}$, and as i runs through the cycle this edge runs through every edge exactly once. A singleton meets at most one vertex of the opposite pair class, and that class contains at most one vertex. Thus two singleton–pair edges use two distinct supports and the two opposite pair vertices. Supports at distance 1 are nonadjacent, for an edge between the corresponding singletons would form a 4-cycle with the cycle edge; supports at distance 2 are nonadjacent, for an edge between the corresponding singletons would form an induced 6-cycle along the longer arc. Pair vertices on distinct edges are nonadjacent, as already recorded.

Suppose the supports are at distance 2: a singleton s at c_0 is adjacent to the pair vertex p on $\{c_2, c_3\}$, and a singleton t at c_2 is adjacent to the pair vertex q on $\{c_4, c_0\}$. The path s, p, c_2, t, q, c_4 is an induced P_6. The consecutive non-edges are the trace restrictions. The distance-2 pairs on the path are sc_2, pt, c_2q and tc_4: s meets only c_0 on the cycle, t meets only its opposite pair vertex q, q meets the cycle only in $\{c_4, c_0\}$, and t meets the cycle only in c_2. The distance-3 pairs are st, pq and c_2c_4, all non-edges by the preceding paragraph or by the cycle. The distance-4 pairs are sq and pc_4, and neither singleton–pair incidence is the opposite one.

Suppose the supports are at distance 1: s at c_0 is adjacent to p on $\{c_2, c_3\}$, and t at c_1 is adjacent to q on $\{c_3, c_4\}$. The path s, p, c_3, q, t, c_1 is an induced P_6, by the same trace restrictions: the only singleton–pair edges among these vertices are sp and tq, the only cycle edges on the path are pc_3 and qc_3, and t is adjacent to c_1 alone on the cycle. Either configuration contradicts the hypothesis that G is P_6-free. Therefore $b \le 1$.

If $b = 0$, then $S = 2n - j - m \le 2n$. If $b = 1$, then $j \ge 1$ because $b \le \min(h, j)$, so $S = 2n + 2 - j - m \le 2n + 1$. In this case $S \le 2n + 1$.

Case 3: no holes. The graph is chordal, and every nonempty induced subgraph has a simplicial vertex. The neighbourhood of a simplicial vertex is a clique. The degree is at most two: four vertices of a clique contain a 4-cycle. A simplicial vertex of a connected chordal graph on at least two vertices is not a cut vertex, and the graph left by deleting it is chordal and strong C_4-free. In the graph present at the deletion, the loss is $L(v) = 2d(v) - 3t(v)$. Degree one gives loss 2. Degree two, with the two neighbours adjacent, gives one triangle and loss 1. Deleting vertices until a single vertex remains, one has $S \le 2(n - 1) = 2n - 2$.

The three cases give the values of $M(n)$. A graph of order $n \ge 11$ cannot realise $k \in \{2, 3\}$ in Case 1, and $k = 0$ forces $G = C_6$. The only remaining graphs in Case 1 are the graphs Γ_h and the graphs with the same first layer and $m \ge 1$. The latter have $S \le 2n + 1$. Cases 2 and 3 give $S \le 2n + 1$ and $S \le 2n - 2$. Thus $M(n) = 2 + 2/n$, attained only by Γ_{n-7}.

At order 10 the only graph in the list with $S > 2 \cdot 10 + 2$ is P, with $S = 30$. At order 9 the only such graph is $P - v$, with $S = 24$, whether it arises as $(k, h) = (3, 0)$ or as $(k, h) = (2, 1)$. At order 8 the only such graph is H_8, with $S = 20$. At order 7 one has $h = 0$ in the equality case of

$k = 1$, so the unique graph with $S = 16$ is Γ_0, while Cases 2 and 3 give at most 15 and 12. Since $8/3 < 3$ and $2 + 2/n < 3$ for $n \geq 11$, the bound $S \leq 3n$ holds for every order $n \geq 2$, with equality only for P. □

Remark 3.7. Karthick and Maffray describe the structure of graphs with no induced P_6 and no induced 4-cycle, and deduce the colouring bound $\chi \leq \lceil 5\omega/4 \rceil$ [12]. Their Theorems 3.2 and 3.3 describe the neighbourhood, on an induced C_5 or C_6, of a vertex outside the cycle. Specialising the 6-cycle description to graphs with no 4-cycle subgraph excludes a consecutive triple, a consecutive quadruple and the full cycle as a support, and leaves an antipodal pair. That is the trace statement at the beginning of Case 1. The constraint on the second layer, the bound $h \leq 1$ when $k \in \{2, 3\}$, and the exact function $M(n)$ are further arguments. They are not consequences of the local support list, and the colouring theorem is not a substitute for the extremal statement under the subgraph prohibition.

Remark 3.8. Brandstädt and Hoàng analysed (P_6, C_4)-free graphs through clique cutsets [2]. In that language C_4 is an induced cycle. A graph in the class with no clique cutset has every induced C_5 dominating, and the presence of an induced C_6 that fails to dominate forces the graph to be the join of a blow-up of the Petersen graph and a clique, possibly empty. Theorem 3.6 is a different statement. The prohibition here is that of a 4-cycle as a subgraph, and the conclusion is the exact maximum of S over every connected graph in the resulting class, including graphs that have clique cutsets. A nontrivial blow-up of the Petersen graph, and a join of the Petersen graph with a nonempty clique, each contain a 4-cycle subgraph, so neither construction enters the class.

Remark 3.9. An alternative proof of $S \leq 3n$, with equality only for the Petersen graph, fixes a vertex of eccentricity at most 2 by Lemma 3.5 and charges the cross edges in the second distance layer. The case division above already gives that inequality and the equality case, so the charging argument is not repeated.

4. Radius four

Lemma 4.1. *Let G be connected, strong C_4-free and P_8-free. If an induced 8-cycle dominates G and $r(G) \geq 4$, then $G = C_8$.*

Proof. Write $C = c_0c_1 \cdots c_7c_0$, with indices modulo 8. For an outside vertex y the trace $A_y = N(y) \cap V(C)$ is nonempty.

The trace A_y is not contained in any three consecutive vertices of C. Suppose $A_y \subseteq \{c_i, c_{i+1}, c_{i+2}\}$. The ends c_i and c_{i+2} are not both in A_y, since y, c_i, c_{i+1}, c_{i+2} would be a 4-cycle. Thus A_y is a singleton or a consecutive pair. A singleton $\{c_i\}$ gives the induced path

$$y, c_i, c_{i+1}, c_{i+2}, c_{i+3}, c_{i+4}, c_{i+5}, c_{i+6}.$$

A consecutive pair $\{c_i, c_{i+1}\}$ gives the induced path

$$y, c_i, c_{i-1}, c_{i-2}, c_{i-3}, c_{i-4}, c_{i-5}, c_{i-6},$$

in which the omitted contact c_{i+1} does not lie on the path. Either path is an induced P_8.

Fix a cycle vertex, say c_0. The vertices at cycle distance greater than two from c_0 are the three consecutive vertices c_3, c_4, c_5. The trace of any outside vertex meets the complementary set $\{c_7, c_0, c_1, c_2, c_6\}$, so every outside vertex is within distance three of c_0. The same holds with c_0 replaced by any vertex of C.

Suppose w lies outside C, and write $A = N(w) \cap V(C)$. No two vertices of A are at cycle distance two, for they would form a 4-cycle with w. A set of vertices of C in which every two distinct vertices are consecutive has size at most two, and every set of size at most two that is a singleton or a consecutive pair is contained in a triple of consecutive vertices. Both possibilities

have been excluded. Therefore A contains two vertices at cycle distance three or four. Rotating, take them to be c_0 and c_d with $d \in \{3, 4\}$.

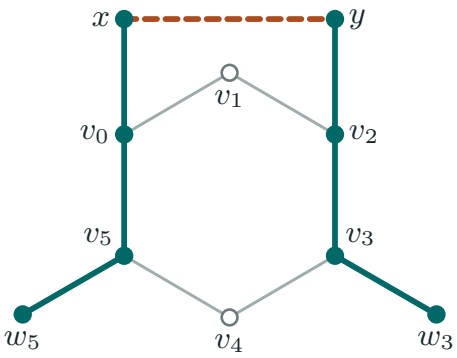
The antipode of c_0 is c_4. If $d = 4$, the path $c_0 w c_4$ has length two. If $d = 3$, then $c_3 \sim c_4$, so $c_0 w c_3 c_4$ has length three. Every other vertex of C is already at cycle distance at most three from c_0. Every outside vertex is within distance three of c_0. Thus the eccentricity of c_0 is at most three, contradicting $r(G) \geq 4$.

Hence C has no vertex outside it, and $G = C_8$. □

Remark 4.2. The proof of Lemma 4.1 does not use a machine enumeration. An independent count, with the labelled cycle fixed, is the script `paper/scripts/c8_configurations.py`. A trace is admissible when the graph formed by the cycle and one vertex of that trace has no 4-cycle subgraph and no induced P_8. Configurations are built by adding admissible traces in increasing bitmask order, joining the new vertex to an arbitrary subset of the outside vertices already chosen, and discarding any graph that acquires a 4-cycle subgraph or an induced P_8. The script finds 32 admissible nonempty traces and 790 configurations, including the bare cycle. By order $8, \ldots, 13$ the counts are $1, 32, 262, 384, 107, 4$. It finds 73 isomorphism classes, and only the bare cycle has radius at least four. That count is not a hypothesis of Theorem 4.3.

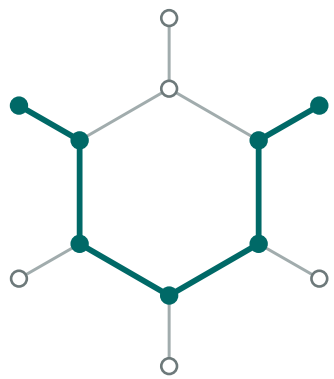
Theorem 4.3. *Let G be connected and strong C_4-free, with $r(G) = 4$ and $p(G) = 7$. Then either $G \cong C_8$, or G is obtained from a 6-cycle $C = v_0 v_1 \cdots v_5$ by the following construction and by no further vertices or edges. For each i there is a nonempty set C_i of vertices outside C. Every vertex of C_i is adjacent to v_i and to no other vertex of C. There is no edge between C_i and C_j for $i \neq j$, and $G[C_i]$ is a matching together with isolated vertices. For each edge of C there is at most one further vertex, adjacent precisely to the two ends of that edge.*

(a) A forbidden cross edge



The dashed edge xy completes an induced P_8.

(b) Equality in $S(G) \leq 2n$



$r = 4, \quad p = 7, \quad S = 24 = 2n.$

FIGURE 2. The radius-four case. (a) If x and y have singleton traces $\{v_0\}$ and $\{v_2\}$ on the dominating 6-cycle, then the hypothetical edge xy, shown dashed, yields the induced path $w_5, v_5, v_0, x, y, v_2, v_3, w_3$. The vertices w_5, w_3 are the distance-four witnesses specified in the proof of Theorem 4.3. (b) Attaching one leaf to each cycle vertex gives $S = 2n$. The highlighted induced path has seven vertices.

Proof. Theorem 2.4 gives $p(G) \geq 7$, which agrees with the hypothesis. Apply Theorem 2.5 with $k = 8$. There is a connected dominating set X such that $G[X]$ is P_6-free or $G[X] \cong C_8$.

If $G[X] \cong C_8$, Lemma 4.1 gives $G = C_8$.

If $G[X]$ is P_6-free, apply Lemma 3.5 inside $G[X]$. A vertex of eccentricity at most two in $G[X]$ has eccentricity at most three in G, because X dominates G, contradicting $r(G) = 4$. Therefore $G[X] = C_6$. Let $C = v_0 \cdots v_5$ be this dominating induced 6-cycle, indices modulo 6, and let $W = V(G) \setminus V(C)$.

Traces. As in the proof of Lemma 3.4, strong C_4-freeness allows only three kinds of trace $N(w) \cap C$: a single vertex, a consecutive pair $\{v_i, v_{i+1}\}$, or an antipodal pair. If some w has trace $\{v_i, v_{i+3}\}$, then v_i is within distance two of every vertex of C, hence within distance three of every vertex of G, a contradiction. So there are no antipodal traces. Let C_i be the set of vertices with trace $\{v_i\}$.

Witnesses. Fix i. Since $r(G) = 4$, some vertex is at distance at least four from v_{i+3}. Every vertex of C is within distance three of v_{i+3}, and a vertex $w \in W$ is at distance at most $1+\min\{d_C(v_{i+3}, x) : x \in N(w) \cap C\}$ from v_{i+3}, which is at most three unless the trace of w is $\{v_i\}$. So there is $w_i \in C_i$ at distance four from v_{i+3}. Every neighbour $z \in W$ of w_i lies in C_i: if z had a neighbour among $v_{i+2}, v_{i+3}, v_{i+4}$, then $d(w_i, v_{i+3}) \leq 3$; if z had the neighbour v_{i-1} or v_{i+1}, then $w_i, v_i, v_{i\pm1}, z$ would be a 4-cycle. So $N(w_i) \subseteq \{v_i\} \cup C_i$.

Edges in W. Figure 2 records the two configurations used below. Let $x, y \in W$ be adjacent.

- $x \in C_0$, $y \in C_{\pm1}$: $x, v_0, v_{\pm1}, y$ is a 4-cycle.
- $x \in C_0$, $y \in C_2$: $w_5, v_5, v_0, x, y, v_2, v_3, w_3$ is an induced P_8.
- $x \in C_0$, $y \in C_3$: $w_1, v_1, v_0, x, y, v_3, v_4, w_4$ is an induced P_8. By symmetry, vertices of C_i and C_j with $i \neq j$ are never adjacent.
- x with trace $\{v_0, v_1\}$ and $y \in C_j$: for $j \in \{0, 1\}$ the vertices v_0, v_1, x, y span a 4-cycle; for $j = 5$ (respectively $j = 2$) the cycle x, v_0, v_5, y (respectively x, v_1, v_2, y) is a 4-cycle; for $j = 3$, $w_5, v_5, v_0, x, y, v_3, v_2, w_2$ is an induced P_8, and $j = 4$ is symmetric.
- x, y with consecutive-pair traces $\{v_0, v_1\}$ and $\{v_j, v_{j+1}\}$: $j = 0$ gives two common neighbours; $j = \pm1$ and $j = \pm2$ give 4-cycles as above; $j = 3$ gives the induced P_8 $w_2, v_2, v_1, x, y, v_4, v_5, w_5$.

In each displayed path the non-edges follow from the traces and from $N(w_i) \subseteq \{v_i\} \cup C_i$. So every edge inside W joins two vertices of the same class C_i, and these edges form a matching. A consecutive-pair vertex is adjacent to no vertex of any C_j and to no other consecutive-pair vertex. Two vertices with the same consecutive-pair trace would have two common neighbours, so each cycle edge carries at most one such vertex, and that vertex is adjacent precisely to the two ends of the edge. Each class C_i is nonempty. This is the stated construction.

Conversely, let G be C_8, or let G be built from $C = v_0 \cdots v_5$ as in the statement. The cycle C_8 has radius 4, induced-path order 7 and no 4-cycle subgraph. Now consider the construction on C. There is no edge between distinct sets C_i, and every vertex of C_i meets C only at v_i. A cycle that meets C_i is therefore contained in $G[C_i \cup \{v_i\}]$. In that subgraph the edges are a star at v_i together with a matching, so the only cycles are the triangles formed by v_i and one matching edge.

Let y be a vertex added on the edge $\{v_j, v_{j+1}\}$, so $N(y) = \{v_j, v_{j+1}\}$. Every cycle through y uses both edges at y, and the rest of the cycle is a v_j–v_{j+1} path avoiding y. A 4-cycle through y is then a path of length two between v_j and v_{j+1} that avoids y: the two ends of the edge have a common neighbour other than y. No vertex of any C_i is adjacent to two vertices of C, and the construction places at most one vertex on each cycle edge. Those two ends therefore have no common neighbour other than the vertex added on that edge. Hence no 4-cycle passes through y. The only cycle that uses no vertex outside C is C itself. Thus G has no 4-cycle subgraph.

Let P be an induced path. An isolated vertex of some $G[C_i]$ has degree one, so it is an end of P if it lies on P. A vertex of a matching edge in C_i has neighbourhood $\{v_i, x'\}$, a clique. It is therefore an end of P whenever P has a third vertex, and P cannot contain both ends of the matching edge together with v_i. A vertex added on a cycle edge has neighbourhood equal to that

edge, again a clique, so it too is an end of any induced path that contains it, and the path cannot contain both ends of the cycle edge. Consequently every induced path of order at least three meets $V(G) \setminus V(C)$ in at most two vertices, both ends of the path. The remaining vertices lie on C, which has no induced path on six vertices. Hence $p(G) \leq 7$. Since each C_i is nonempty, one may choose $x \in C_0$ and $y \in C_2$. The path $x, v_0, v_5, v_4, v_3, v_2, y$ is induced: x and y meet the path only at v_0 and v_2, and C contributes no chord. Thus $p(G) = 7$.

There is no edge between distinct sets C_i, and a vertex added on a cycle edge has degree two on that edge, so it does not shorten distances along C. The distance from v_i to v_{i+3} is therefore three. A vertex of C_{i+3} is at distance four from v_i, and the eccentricity of v_i is at least four. It is at most four: every vertex of C_j is at distance $d_C(v_i, v_j) + 1 \leq 4$ from v_i, and every vertex added on a cycle edge is within distance four of v_i. A vertex of C_i is at distance five from a vertex of C_{i+3}. A vertex y added on the edge $\{v_j, v_{j+1}\}$ has only those two neighbours. The distances from v_j and from v_{j+1} to v_{j+3} are 3 and 2, neither neighbour of y is adjacent to v_{j+3}, and $y, v_{j+1}, v_{j+2}, v_{j+3}$ is a path of length three, so a vertex of C_{j+3} is at distance four from y. Every eccentricity is at least four, and $r(G) = 4$. □

Corollary 4.4. *Let G be connected and strong C_4-free with $r(G) = 4$ and $p(G) = 7$. In the construction of Theorem 4.3, let b be the number of vertices added on edges of the 6-cycle and let m be the number of matching edges in the sets C_i. Then $S(G) = 2n - b - m$. In particular $S(G) \leq 2n$, so $\ell(G) \leq 2$, and* RAD(4) *holds.*

Proof. For $G \cong C_8$ one has $n = 8$, $e = 8$, $T = 0$ and $S = 16 = 2n$, which is the formula with $b = m = 0$. In the construction, write $a = \sum_i |C_i|$. Then $n = 6 + a + b$. The edges are the six cycle edges, a edges from the sets C_i to the cycle, m matching edges and $2b$ edges incident with the vertices added on cycle edges, so $e = 6 + a + 2b + m$. Each matching edge forms one triangle with its cycle vertex, and each added vertex forms one triangle with its cycle edge, so $T = b + m$. Lemma 2.1(3) gives $S = 2e - 3T = 2n - b - m$. □

Corollary 4.5. *Let G be connected and strong C_4-free with $r(G) = 4$ and $p(G) = 7$. Then $S(G) = 2n$ if and only if $G \cong C_8$, or G is obtained from C_6 by attaching at least one leaf to each cycle vertex.*

Proof. By Corollary 4.4, $S = 2n$ if and only if $G \cong C_8$ or, in the construction, $b = m = 0$. In the latter case every vertex outside the cycle is a leaf, and each C_i is nonempty, so each cycle vertex has at least one leaf. The converse is the case $b = m = 0$ of the construction, together with C_8, both of which have $S = 2n$ and satisfy the numerical hypotheses by Theorem 4.3. □

Remark 4.6. If G is connected and strong C_4-free with $r(G) = 4$, $p(G) \geq 8$ and $\ell(G) \leq 4$, then $p(G) \geq 8 = r(G) + 4 \geq r(G) + \ell(G)$. This comparison is not a structural description of G.

5. Triangle-free graphs of induced-path order at most six

In a triangle-free graph, a 4-cycle subgraph is an induced 4-cycle, and every clique has order at most two. The next classification is Lemma 3.4 of Chen and Xu [5]. Their family $\mathfrak{F} = \{F_0, F_0', F_1, \ldots, F_{12}\}$ is defined by the constructions in the proofs of their Lemmas 3.1–3.3; F_5 is the Petersen graph, and the parameters t and t' attached to F_2 and F_0' are positive integers. The edge counts below use those constructions rather than the accompanying figure.

Lemma 5.1 (Chen and Xu [5, Lemma 3.4])**.** *Let G be a connected (P_7, C_4, C_3)-free graph with no clique cutset. If G is not complete, then $G \in \mathfrak{F}$.*

The members of $\mathfrak{F}$ admit the following descriptions, taken from those proofs.

- F_1 is C_5. The graph F_2 is obtained from a 5-cycle $v_1 \cdots v_5$ by adding stable sets A_1 and A_3 of equal positive order t, with a perfect matching between them, each vertex of A_1 adjacent only to v_1 and its match, and each vertex of A_3 adjacent only to v_3 and its match.

- On the same 5-cycle, F_3 has three further vertices inducing the path $a_3a_1a_4$, with a_i adjacent to v_i. The graph F_4 has four further vertices inducing the path $a_5a_3a_1a_4$, with the same spokes. The graph F_5 is the Petersen graph.
- F_0 is C_6. The graph F_0' is obtained from a 6-cycle $v_1 \cdots v_6$ by adding stable sets Y_1 and Y_4 of equal positive order t', with a perfect matching between them, each vertex of Y_1 adjacent only to v_1 and its match, and each vertex of Y_4 adjacent only to v_4 and its match.
- F_6 is C_7. On a 7-cycle $v_1 \cdots v_7$, write x_i for a vertex with neighbourhood $\{v_i, v_{i+3}\}$ on the cycle. Then F_7 is the cycle plus x_1; F_8 is the cycle plus nonadjacent x_1, x_2; F_9 is F_8 plus a vertex adjacent only to x_1 and x_2; F_{10} is the cycle plus nonadjacent x_1, x_5; F_{11} is the cycle plus a stable set $\{x_1, x_2, x_5\}$; and F_{12} is F_{11} plus a vertex adjacent only to x_1 and x_2.

Chen and Xu record that F_0, F_1, the graph F_2 with $t = 1$, F_3 and F_4 are induced subgraphs of the Petersen graph, and that $F_6, \ldots, F_{11}$ are induced subgraphs of F_{12}. The graphs $F_6, \ldots, F_{12}$ are the extensions of C_7 by vertices of cycle-distance three. The graphs F_2 and F_0' are the matching extensions of one pair of opposite vertices on C_5 and on C_6.

Lemma 5.2. *Every graph in $\mathfrak{F}$ other than the Petersen graph satisfies $2e \le 3n - 3$. The Petersen graph satisfies $2e = 3n$.*

Proof. The constructions give the following orders and sizes. For F_2 and F_0' the parameters t, t' are at least one.

Graph	n	e	$2e$	$3n - 2e$
$F_1 = C_5$	5	5	10	5
F_2	$5 + 2t$	$5 + 3t$	$10 + 6t$	5
F_3	8	10	20	4
F_4	9	12	24	3
F_5	10	15	30	0
$F_0 = C_6$	6	6	12	6
F_0'	$6 + 2t'$	$6 + 3t'$	$12 + 6t'$	6
$F_6 = C_7$	7	7	14	7
F_7	8	9	18	6
F_8, F_{10}	9	11	22	5
F_9, F_{11}	10	13	26	4
F_{12}	11	15	30	3

The five edges of F_3 beyond C_5 are the three spokes and the two edges of $a_3a_1a_4$. The seven edges of F_4 beyond C_5 are the four spokes and the three edges of $a_5a_3a_1a_4$. Each vertex x_i contributes two edges, and each vertex placed on a pair x_ix_j contributes two further edges. The Petersen graph is the only row with difference zero. □

Lemma 5.3. *Let F be connected and strong C_4-free, and let s be a vertex at which no induced P_4 starts. Write $m = e(F[N(s)])$. Then $S(F) \le 2(|F| - 1) - m$.*

Proof. A vertex at distance 3 from s would end a geodesic, hence an induced P_4, starting at s. So $V = \{s\} \cup N_1 \cup N_2$ with N_i the i-th distance layer. Each $y \in N_2$ has a unique parent $a \in N_1$. A neighbour $z \in N_2$ of y with a different parent would give the induced path $sayz$. So y has at most one neighbour in N_2, a sibling (two siblings would be common neighbours of a and y), and $N(y)$ is a clique; thus $\lambda(y) = 1$. For $a \in N_1$ we have $N(a) \subseteq \{s, \bar{a}\} \cup C_a$, where C_a is the set of children of a and $\bar{a}$ is the neighbour of a in N_1, if any. Since s and $\bar{a}$ lie in one unit, $\lambda(a) \le 1 + |C_a|$. Finally $\lambda(s) = |N_1| - m$. Summing, $S \le (|N_1| - m) + (|N_1| + |N_2|) + |N_2| = 2(|F| - 1) - m$. □

Theorem 5.4. *Let G be a connected triangle-free strong C_4-free P_7-free graph. Then $2e(G) \leq 3|V(G)|$, with equality if and only if G is the Petersen graph. In particular every connected triangle-free strong C_4-free graph of induced-path order at most 6 satisfies GEN_3, and so does every connected bipartite strong C_4-free graph of induced-path order at most 6. A bipartite graph in the class has strict inequality.*

Proof. The proof is by induction on $n = |V(G)|$. The graph K_2 has $2e = 2 < 3n$. Suppose $n \geq 3$, and write $S = 2e$, which is legitimate because G is triangle-free.

Suppose first that v is a cut vertex. For each component C of $G - v$ let ρ_C be the largest order of an induced path in $G[C \cup \{v\}]$ that starts at v. Paths from two components form an induced path of G on $\rho_C + \rho_{C'} - 1 \leq 6$ vertices, so some component C_1 has $\rho_{C_1} \leq 3$. Let $G_1 = G[C_1 \cup \{v\}]$ and $G_2 = G - C_1$, with orders n_1 and n_2. Then $n = n_1 + n_2 - 1$ and $n_1 \geq 2$. Lemma 2.3 gives $S(G) = S(G_1) + S(G_2)$. Lemma 5.3 gives $S(G_1) \leq 2(n_1 - 1)$. The graph G_2 is a smaller connected triangle-free strong C_4-free P_7-free graph, so the inductive hypothesis gives $S(G_2) \leq 3n_2$. Therefore

$$S(G) \leq 2(n_1 - 1) + 3n_2 = 3n - (n_1 - 1) < 3n.$$

If G has a cut vertex, this estimate gives the theorem, and equality does not hold. Assume for the remainder of the argument that G has no cut vertex.

Suppose that $\{u, v\}$ is a clique cutset of order two. Then uv is an edge. Since neither vertex is a cut vertex, every component of $G - \{u, v\}$ is adjacent to both u and v, and there are at least two such components. A shortest u–v path through one component has at least four edges: one internal vertex would form a triangle with uv, and two internal vertices would form a 4-cycle. In two components, choose shortest paths, and on each path let x, y, z be the three vertices immediately preceding v, in that order from the u-side. The same choice in the second component produces x', y', z'. The path x, y, z, v, z', y', x' is an induced P_7. Indeed, each side is a subpath of a shortest path, so it has no chord, and there is no edge between $\{x, y, z\}$ and $\{x', y', z'\}$ because those sets lie in different components of $G - \{u, v\}$. This contradicts $p(G) \leq 6$.

Thus G has no clique cutset. A triangle-free graph has no 4-cycle subgraph if and only if it has no induced 4-cycle, so Lemma 5.1 applies: G is complete or belongs to $\mathfrak{F}$. The only complete triangle-free graph on at least two vertices is K_2, already strict. Lemma 5.2 gives $2e \leq 3n$ for every graph in $\mathfrak{F}$, with equality only for the Petersen graph.

A bipartite graph is triangle-free. The Petersen graph is not bipartite, so the inequality is strict for every connected bipartite strong C_4-free P_7-free graph. □

Corollary 5.5. *Let G be connected and strong C_4-free with $r(G) = 3$, $p(G) = 6$ and $\ell(G) > 3$. Then G contains a triangle.*

Proof. If G were triangle-free, Theorem 5.4 would give $S(G) \leq 3n$. □

Lemma 5.6. *Let G be a connected strong C_4-free P_7-free graph on $n \geq 2$ vertices. Then $e(G) \leq 2n - 3$. Consequently $S(G) \leq 4n - 6 - 3T(G)$, so $\ell(G) < 4$. If G contains a triangle, then $\ell(G) \leq 4 - 9/n$.*

Proof. The proof of the edge bound is by induction on n. For $n = 2$ one has $G = K_2$ and $e = 1 = 2n - 3$. A copy of K_4 contains a 4-cycle, so $\omega(G) \leq 3$. A diamond contains a 4-cycle, so G is $(P_7, C_4, \text{diamond})$-free in the sense of Chen, Wu and Xu [6].

Suppose K is a clique cutset, necessarily of order $t \in \{1, 2, 3\}$, and let $G - K$ have $k \geq 2$ components. For each component C let G_C be the subgraph induced by $C \cup K$, of order $n_C < n$. Each G_C is connected, strong C_4-free and P_7-free. The edges inside K are counted k times, and $n = \sum n_C - (k-1)t$, so

$$e(G) = \sum_C e(G_C) - (k-1)\binom{t}{2}.$$

The inductive hypothesis gives $e(G_C) \le 2n_C - 3$, and therefore

$$\begin{aligned} e(G) &\le \sum_C (2n_C - 3) - (k-1)\binom{t}{2} \\ &= 2n + (k-1)\Big(2t - \binom{t}{2}\Big) - 3k \\ &= 2n - 3 + (k-1)\Big(2t - 3 - \binom{t}{2}\Big). \end{aligned}$$

The factor $2t - 3 - \binom{t}{2}$ equals -1 if $t = 1$ and equals 0 if $t = 2$ or $t = 3$. Hence $e(G) \le 2n - 3$.

Suppose G has no clique cutset. Theorem 1.1 of [6] states that a connected $(P_7, C_4, \text{diamond})$-free graph with no clique cutset is the Petersen graph, or the graph F defined below, or else has minimum degree at most $\max\{2, \omega - 1\}$. Here $\omega \le 3$, so the last quantity is at most 2. The Petersen graph has $n = 10$ and $e = 15$, and $2n - 3 = 17$. The graph F is obtained from a 7-cycle $v_1 \cdots v_7$ by adding an independent set $\{y_1, y_2, y_3\}$ with

$$N(y_1) = \{v_1, v_4, v_5\}, \qquad N(y_2) = \{v_2, v_5, v_6\}, \qquad N(y_3) = \{v_3, v_6, v_7\}.$$

Thus F has $n = 10$ and $e = 7 + 9 = 16 \le 17$. If instead some vertex v has degree at most 2, then v is not a cut vertex, so $G - v$ is connected. For $n \ge 3$ the inductive hypothesis gives $e(G - v) \le 2(n-1) - 3$, and $e(G) \le 2n - 5 + d(v) \le 2n - 3$.

Lemma 2.1(3) now gives $S = 2e - 3T \le 4n - 6 - 3T \le 4n - 6$, so $\ell \le 4 - 6/n < 4$. If $T \ge 1$, then $S \le 4n - 9$ and $\ell \le 4 - 9/n$. □

Corollary 5.7. *Let G be connected and strong C_4-free with $r(G) = 3$, $p(G) = 6$ and $\ell(G) > 3$. Then G contains a triangle and $3 < \ell(G) \le 4 - 9/n$. In particular $\ell(G) \ne 4$, and no graph of order at most 9 lies in this cell.*

Proof. Corollary 5.5 supplies a triangle. Since $p(G) = 6$, the graph is P_7-free, and Lemma 5.6 gives $\ell \le 4 - 9/n$. For $n \le 9$ one has $4 - 9/n \le 3$. □

Remark 5.8. The degree bound in Theorem 1.1 of Chen, Wu and Xu does not by itself yield $S \le 3n$. Lemma 5.6 is the edge estimate that follows from that theorem once clique cutsets of order 1, 2 and 3 are restored and the two exceptional graphs are counted. The resulting bound is $\ell < 4$, not $S \le 3n$. The case of GEN_3 in which triangles are present remains open. Appendix A records a conditional reduction for that case.

Theorem 5.9 (Theorem A)**.** *Let G be connected and strong C_4-free with $n \ge 2$. Then $p(G) \ge r(G) + \ell(G)$ whenever $\ell(G) \le 3$, whenever $r(G) \le 2$ and $\ell(G) \le 4$, whenever $r(G) = 4$ and $\ell(G) \le 4$, and whenever $r(G) \ge 5$ and $\ell(G) \le 4$. If $r(G) = 4$ and $p(G) = 7$, then $\ell(G) \le 2$.*

Proof. Write $r = r(G)$ and $c = \lceil \ell(G) \rceil$. The last sentence is Corollary 4.4. For the inequality we need $p \ge r + c$ in the listed ranges.

- $r = 1$: Lemma 3.1.
- $r = 2$, $c \le 2$: Lemma 3.2.
- $r = 2$, $c = 3$: if $p \le 4$, then Lemma 3.3 gives $\ell \le 2$, a contradiction. So $p \ge 5$.
- $r = 2$, $c = 4$: if $p \le 5$, then Theorem 3.6 gives $\ell \le 3$, a contradiction. So $p \ge 6$.
- $r = 3$, $c \le 2$: Theorem 2.4 gives $p \ge 5$.
- $r = 3$, $c = 3$: if $p \le 5$, then by Lemma 3.5 and $r = 3$ we have $G = C_6$, with $\ell = 2$, a contradiction. So $p \ge 6$. Theorem 3.6 gives $\ell \le 3$ for every P_6-free graph in the class, which does not rule out $c = 3$.
- $r \ge 4$, $c \le 3$: Theorem 2.4 gives $p \ge 2r - 1 \ge r + 3$.
- $r = 4$ and $\ell \le 4$: Theorem 2.4 gives $p \ge 7$. If $p = 7$, Corollary 4.4 gives $\ell \le 2$, so $p \ge r + \ell$. If $p \ge 8$, Remark 4.6 gives the same inequality.

- $r \geq 5$ and $\ell \leq 4$: Theorem 2.4 gives $p \geq 2r-1 \geq r+4 \geq r+c$.

□

Equality in REAL-133 occurs for C_5 ($r=2$, $\ell=2$, $p=4$), the Petersen graph ($r=2$, $\ell=3$, $p=5$) and C_6 ($r=3$, $\ell=2$, $p=5$). We do not characterise all equality cases.

6. The reduction and the residual cases

For integers $K \geq 3$ and $r \geq 4$ consider the following statements.

GEN_K: every connected strong C_4-free graph G with $p(G) \leq 2K$ satisfies $S(G) \leq Kn$.
$\mathrm{RAD}(r)$: every connected strong C_4-free graph with $r(G) = r$ and $p(G) = 2r-1$ satisfies $\ell(G) \leq r-1$.

Corollary 4.4 is the case $r = 4$ of $\mathrm{RAD}(r)$, in the stronger form $\ell \leq 2$.

Corollary 6.1 (Theorem B)**.** *Let G be a connected strong C_4-free graph that violates Conjecture 1.1. Then $\ell(G) > 3$. Put $K = \lceil \ell(G) \rceil - 1 \geq 3$. Then either*

(a) *$r(G) \leq K$ and $p(G) \leq r(G) + K \leq 2K$, so G violates GEN_K; or*
(b) *$r(G) = K+1 \geq 4$ and $p(G) = 2r(G)-1$, and $\ell(G) > r(G)-1$, so G violates $\mathrm{RAD}(K+1)$.*

Proof. Write $r = r(G)$, $p = p(G)$ and $c = \lceil \ell(G) \rceil = K+1$. Then $\ell > 3$ by Theorem 5.9. A violation means $p \leq r+c-1 = r+K$. Theorem 2.4 gives $2r-1 \leq p \leq r+K$, so $r \leq K+1$. If $r \leq K$, then $p \leq r+K \leq 2K$, and $S > Kn$ because $\ell > K$. If $r = K+1$, then $2r-1 \leq p \leq 2K+1 = 2r-1$, and $\ell > K = r-1$. □

Since GEN_K ignores the radius, it is stronger than alternative (a). The statement GEN_K might fail for large K even if REAL-133 holds; see Remark 8.5.

Proposition 6.2. *For $k \geq 3$ let θ_k consist of two vertices x, y joined by k internally disjoint paths xa_ib_iy. Then θ_k is strong C_4-free, $n = 2k+2$, $r = 3$, $p = 6$ and $S = 6k = 3n-6$. Thus $p - r - \ell = 6/n \to 0$.*

Proof. Every cycle consists of two complete branches and so has length 6; in particular there is no 4-cycle. There are no triangles, so $S = 2e = 6k$. Every vertex has eccentricity 3: $d(x,y) = 3$ and $d(a_i, b_j) = 3$ for $i \neq j$. The path $a_jxa_ib_iyb_l$ (with j, i, l distinct) is induced. Conversely, let P be an induced path. If P avoids x and y, it lies inside one edge a_ib_i. If P contains x but not y, then x has at most two neighbours on P, each followed on P by at most its b-vertex, so $|P| \leq 5$. If P contains x and y, then every a_j on P is adjacent on P to x and every b_j to y, at most one branch lies entirely on P, and so P consists of x, y, at most one full branch and at most one further neighbour of each of x and y. Hence $|P| \leq 6$. □

Thus no argument with a uniform positive margin in ℓ can prove GEN_3, already among graphs of radius 3. The resulting division of REAL-133 is recorded once, in Table 1.

Let $3 < \ell \leq 4$, and suppose G violates REAL-133. Theorem 6.1 gives $K = 3$, so either $r \leq 3$ and $p \leq 6$, or $r = 4$ and $p = 7$ with $\ell > 3$. The second alternative contradicts Corollary 4.4. If $r \leq 2$, Theorem 5.9 applies. Thus $r = 3$ and $p \leq 6$. Theorem 2.4 gives $p \geq 5$. If $p \leq 5$, Theorem 3.6 gives $\ell \leq 3$. Hence $p = 6$. If instead $p \geq 7$, then $p \geq 7 \geq r + \ell$. For $r \geq 5$ and $\ell \leq 4$, Theorem 2.4 gives $p \geq 2r-1 \geq r+4$, so REAL-133 holds with no further structural work. Every counterexample in the range $\ell \leq 4$ therefore satisfies $r = 3$ and $p = 6$. Corollary 5.7 sharpens the inequality to $3 < \ell \leq 4 - 9/n$ and shows that the graph contains a triangle. In particular $\ell = 4$ does not occur. The parts of the argument that are not the Erdős–Saks–Sós bound are Theorems 3.6, 4.3 and 5.4, together with the edge bound of Lemma 5.6. The inequality GEN_3 for graphs that contain a triangle would close the remaining cell. It is not proved in this paper; see Appendix A.

Lemma 6.3. *Let G be connected, bipartite and strong C_4-free, with $r(G) = 2$. Then G is a tree, $\ell(G) = 2 - 2/n$ and $p(G) \le 5$.*

Proof. Let s be a centre, and let A be the colour class of s and B the other class. Every vertex of B lies at odd distance from s, hence at distance 1, so $B \subseteq N(s)$. Every vertex $x \in A \setminus \{s\}$ lies at distance 2 from s, so x has a neighbour in B. Two such neighbours would form a 4-cycle with s. Thus x has exactly one neighbour. There is no edge inside A or inside B, and the number of edges is $|B| + |A| - 1 = n - 1$. A connected graph with $n - 1$ edges is a tree. There are no triangles, so $S = 2e = 2n - 2$ and $\ell = 2 - 2/n$. Every vertex of $A \setminus \{s\}$ is a leaf. An induced path contains at most two leaves, and the unique path between them has the form leaf, neighbour in B, s, neighbour in B, leaf. Hence $p(G) \le 5$. □

Lemma 6.4. *Let G be a connected bipartite strong C_4-free P_8-free graph. Then $2e(G) \le 4n$, so $\ell(G) \le 4$.*

Proof. The bound holds for $n = 1$. Let $n \ge 2$. A bipartite strong C_4-free graph has girth at least 5. Theorem 1 of Chudnovsky and Stacho [4] states that a connected P_8-free graph of girth at least 5 is the Petersen graph, the Heawood graph, the graph obtained by contracting one edge of the Heawood graph, or else has a vertex of degree at most 2. The Petersen graph is not bipartite. The Heawood graph is bipartite of girth 6; contracting an edge that lies on a 6-cycle produces a 5-cycle, so the contraction is not bipartite. The Heawood graph has $n = 14$ and $e = 21$, and $2e = 42 \le 56 = 4n$.

Thus a bipartite graph in the class that is not the Heawood graph has a vertex v of degree at most 2. Every component H of $G - v$ is connected, bipartite, strong C_4-free and P_8-free. The inductive hypothesis gives $2e(H) \le 4|H|$, so

$$2e(G) = 2e(G - v) + 2d(v) \le 4(n - 1) + 4 = 4n,$$

because $d(v) \le 2$. Since G is bipartite, $S = 2e$ and $\ell \le 4$. □

Lemma 6.3 excludes the pair $(r, p) = (2, 7)$ for every connected bipartite strong C_4-free graph: the induced-path order is at most 5, and $\ell < 2$. Lemma 6.4 excludes $(r, p) = (3, 7)$ from the range $5 < \ell \le 6$, because a graph with $p = 7$ is P_8-free and then $\ell \le 4$. Lemma 5.6 excludes $(2, 6)$ and $(3, 6)$ for every connected strong C_4-free graph, bipartite or not: $p \le 6$ forces $\ell < 4$.

7. Rooted bounds of orders three and four

For a vertex s of a connected graph F let $p_s(F)$ be the largest order of an induced path starting at s. Write $D = d_F(s)$ and $m = e(F[N(s)])$.

Definition 7.1. For an integer $q \ge 3$, the rooted bound ROOT(q) asserts that every connected strong C_4-free graph F and every vertex s with $p_s(F) \le q$ satisfy

$$S(F) \le (q - 1)(|F| - 1) - (q - 3)D - m.$$

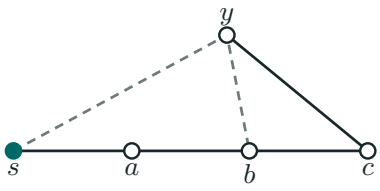
The hypothesis $p_s(F) \le q$ belongs in the statement. Without it the displayed inequality is false: if $q = 3$ and $F = C_5$, then $S(F) = 10$ and $m = 0$, while $2(|F| - 1) = 8$. Lemma 5.3, already proved because the cut-vertex induction of Theorem 5.4 uses it, is the case $q = 3$.

Theorem 7.2. *Let F be connected and strong C_4-free, and let $s \in V(F)$ satisfy $p_s(F) \le 4$. Write $D = d(s)$ and $m = e(F[N(s)])$. Then*

$$S(F) \le 3(n - 1) - D - m. \tag{2}$$

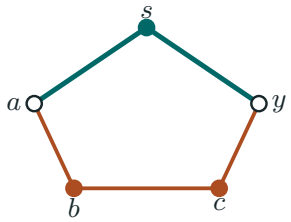
In particular ROOT(4) *holds. Equality holds for C_5 at any root, and for a graph formed by $k \ge 1$ copies of C_5 that meet only at the root.*

(a) A longest induced path from s



Each extra neighbour y sees s or b (dashed edges). The edge ay would create a 4-cycle.

(b) Deleting two degree-2 endpoints



If $d(b) = d(c) = 2$, delete $U = \{b, c\}$. The path a, s, y survives and $D_G(U) = 6$.

FIGURE 3. Deletion at the end of a longest induced path of order four starting at s. (a) Each neighbor $y \neq b$ of c must be adjacent to s or b. Dashed lines indicate alternatives. (b) If neither terminal vertex c nor b can be deleted with loss at most three, the indicated 5-cycle is induced and both have degree two. The deletion of $\{b, c\}$ has loss six and preserves connectivity.

Proof. The proof is by induction on n. If $p_s(F) \leq 3$, Lemma 5.3 gives $S \leq 2(n-1) - m$. Since $D \leq n - 1$, the right-hand side is at most $3(n-1) - D - m$.

Now suppose $p_s(F) = 4$, and let s, a, b, c be an induced path. The endpoint c is not a cut vertex: a neighbour of c in a component of $F - c$ that does not contain s would extend the path. Let $y \in N(c) \setminus \{b\}$. The path s, a, b, c, y is not induced, so y meets at least one of s, a, b. The edge ya would give the 4-cycle a, b, c, y. Thus y is adjacent to s or to b. The vertices s and c have at most one common neighbour, and so do b and c, hence $d(c) \leq 3$. If $d(c) = 3$, then c has two neighbours besides b, and one of them is adjacent to b. So c lies in a triangle, $t(c) \geq 1$ and $L(c) = 2d(c) - 3t(c) \leq 3$. Figure 3 shows the two deletions.

If $L(c) \leq 3$, delete c. The graph $F - c$ is connected, and $c \notin N[s]$, so D and m are unchanged. The inductive hypothesis gives $S(F - c) \leq 3(n-2) - D - m$, and therefore $S(F) \leq 3(n-2) - D - m + 3 = 3(n-1) - D - m$.

The remaining case is $d(c) = 2$ and $t(c) = 0$, so $L(c) = 4$. Let y be the neighbour of c other than b. Then y is not adjacent to b, hence ys is an edge, and s, a, b, c, y is an induced 5-cycle: every chord is excluded by the induced path s, a, b, c, by $t(c) = 0$, or by the 4-cycle that ay would create. The vertex b is the endpoint of the induced path s, y, c, b. If $L(b) \leq 3$, delete b and apply the inductive hypothesis as above.

Otherwise the same endpoint analysis gives $d(b) = 2$ and $t(b) = 0$. Delete $U = \{b, c\}$. A path that used the edges ab, bc, cy can be replaced along a, s, y, so $F - U$ is connected. The deleted edges are ab, bc and cy, and no triangle is deleted. Since $S = 2e - 3T$, one has $D_F(U) = 6 = 3|U|$. The vertices b and c lie outside $N[s]$, so D and m are unchanged. Induction gives $S(F - U) \leq 3(n-3) - D - m$, and $S(F) \leq 3(n-3) - D - m + 6 = 3(n-1) - D - m$.

For the equality examples, root C_5 at any vertex. Then $n = 5$, $D = 2$, $m = 0$ and $S = 10 = 3(n-1) - D - m$. If k copies of C_5 meet only at the root, then $n = 1 + 4k$, $D = 2k$ and $m = 0$, because the two neighbours of the root on each copy are nonadjacent, and $S = 10k = 3(n-1) - D - m$. □

Proposition 7.3. *Let F be connected and strong C_4-free, and let s satisfy $p_s(F) = 4$ and $\mathrm{ecc}(s) = 3$. Then $S(F) \leq 3(n-1) - D - m - 2$. Equality holds for P_4 rooted at an endpoint.*

Proof. The proof is by induction on n for instances with $p_s = 4$ and $\mathrm{ecc}(s) = 3$. Fix a vertex z at distance 3 from s. Whenever an endpoint other than z exists, the deletion in the proof of Theorem 7.2 applies to it. In the exceptional 5-cycle the vertices b and c are at distance 2 from s, so the deletion leaves z. The remaining graph is connected and strong C_4-free, D and m are

unchanged, and distances do not decrease. Thus $d(s,z) \geq 3$. Since $p_s \leq 4$, no distance exceeds 3, so $p_s = 4$ and the eccentricity of s is still 3. A deletion of one vertex of loss at most 3 contributes 3, and the inductive bound on $n-1$ vertices contributes $3(n-2) - D - m - 2$, which sum to $3(n-1) - D - m - 2$. Deleting $\{b, c\}$ contributes 6 against $3(n-3) - D - m - 2$.

If z is the unique endpoint, delete z. Then $p_s(F - z) \leq 3$. An extra neighbour of z is not adjacent to s, for that would give $d(s,z) \leq 2$. By the endpoint analysis it is adjacent to the predecessor of z and forms a triangle, so $L(z) \leq 2$. Lemma 5.3 gives $S(F - z) \leq 2(n-2) - m$, hence $S(F) \leq 2(n-1) - m$. A geodesic from s to z has at least two vertices outside $N[s]$, so $n - 1 - D \geq 2$ and $2(n-1) - m \leq 3(n-1) - D - m - 2$.

For P_4 rooted at an endpoint one has $n = 4$, $D = 1$, $m = 0$ and $S = 6$, and the strengthened bound is $3 \cdot 3 - 1 - 2 = 6$. □

Proposition 7.4. *Let G be a counterexample to* GEN_4 *of minimum order. Then G is 2-connected. This does not decide* GEN_4.

Proof. Such a graph is connected and strong C_4-free, with $p(G) \leq 8$ and $S(G) \geq 4n + 1$. Let v be a cut vertex, and let G_1, G_2 be the graphs of Lemma 2.3. Write ρ_i for the largest order of an induced path of G_i that starts at v. Paths from the two sides form an induced path of G on $\rho_1 + \rho_2 - 1 \leq 8$ vertices, so some side, say G_1, has $\rho_1 \leq 4$. Theorem 7.2 gives $S(G_1) \leq 3(n_1 - 1)$. The graph G_2 is a smaller connected strong C_4-free graph of path order at most 8, so minimality gives $S(G_2) \leq 4n_2$. Therefore $S(G) \leq 3(n_1 - 1) + 4n_2$. Since $n = n_1 + n_2 - 1$,

$$3(n_1 - 1) + 4n_2 = 4n - n_1 + 1.$$

The hypothesis $n_1 \geq 2$ gives $4n - n_1 + 1 \leq 4n - 1$, so $S(G) < 4n$, contradicting $S(G) \geq 4n + 1$. □

Remark 7.5. The same bound has a distance-layer proof. Let $N_i = \{v : d(s,v) = i\}$. There is no vertex of N_4: a geodesic from s to such a vertex is an induced path on five vertices. Each vertex of N_2 has a unique parent in N_1. The argument below gives

$$S(F) \leq 2D - m + 3|N_2| + 2|N_3|. \tag{3}$$

Since $n - 1 = D + |N_2| + |N_3|$, the right-hand side equals $3(n-1) - D - m - |N_3|$, and (2) follows.

Take $z \in N_3$ and a path s, a, c, z with $a \in N_1$ and $c \in N_2$. If $y \neq c$ is another neighbour of z, then y is not adjacent to s, and y is not adjacent to a, for a, c, z, y would be a 4-cycle. The path s, a, c, z, y is not induced, so cy is an edge. Thus c and z have at most one common neighbour, $d(z) \leq 2$ and $N(z)$ is a clique. Hence $L(z) \leq 2$. Delete the vertices of N_3 one at a time. Each deletion has loss at most 2, and D and m stay unchanged.

Let $H = F[N_2]$. If a vertex $c \in H$ meets H only in the class of its parent, then its degree in H is at most one. If cd is an edge of H with d in a different class, and b is the parent of d, then every other H-neighbour y of c fails to be adjacent to b, for b, d, c, y would be a 4-cycle, and the path s, b, d, c, y forces the edge dy. Since c and d have at most one common neighbour, $N_H(c)$ is a clique. In every case $\lambda_H \leq 1$, so $S(H) \leq |N_2|$.

Let h be the number of edges of H with both ends in one parent class. After N_3 is deleted, the remaining graph has $D + m + |N_2| + e(H)$ edges and $m + h + T(H)$ triangles: the m edges at s, the h triangles formed by a parent and two of its children, and the triangles inside H. No other triangle meets N_1 and N_2, because a vertex of N_2 has only one parent. Thus, writing $S(H) = 2e(H) - 3T(H)$,

$$S(F - N_3) = 2D - m + 2|N_2| + S(H) - 3h \leq 2D - m + 3|N_2|.$$

Adding the deletion loss of N_3, at most $2|N_3|$, gives (3).

For $q \geq 5$ the bound $\mathrm{ROOT}(q)$ is not proved for arbitrary strong C_4-free graphs. Section 10 records the limitation. The next section proves every index for graphs whose blocks are bipartite

or cycles. Conjecture 10.2 would imply ROOT(q) for the same q, by the attachment in Section 10. That implication is not a proof of the conjecture, and for $q \geq 5$ it does not prove ROOT(q).

8. Rooted bounds for every index

Let $\mathcal{B}$ be the class of connected strong C_4-free graphs in which every block is bipartite or a cycle. Edges are blocks, and cycles may be odd. The class contains every connected strong C_4-free bipartite graph and every connected strong C_4-free cactus. Theorem 8.2 keeps this block hypothesis. The graph $K_{2,3}$ does not show that the hypothesis is necessary. Its only block is itself, and that block is bipartite, but $K_{2,3}$ contains a 4-cycle subgraph. Rooted in its part of size two it has $p_s = 3$ and $S = 12 > 2(n-1) - m = 8$. The example tests the failure of ROOT(3) after the strong C_4-free hypothesis is removed. It does not test the block restriction inside the strong C_4-free class.

A rooted graph (G, s) with $p_s(G) = q$ and $\mathrm{ecc}(s) = q - 1$ is called *terminal*. For a terminal instance put $R = n - q + 1 - D \geq 0$, the number of vertices outside $N[s]$ and outside a root-to-farthest-vertex geodesic. Set $c_2 = 0$, $c_3 = 2$, $c_4 = 3$ and $c_q = 2\lfloor (q-1)/2 \rfloor$ for $q \geq 5$.

Lemma 8.1. *Let G be connected and strong C_4-free, and let P be a longest induced path starting at a vertex s, of order q. Let x be the other end of P, and suppose that the block B containing x is bipartite. Let Q be the subpath of P formed by the vertices of P that lie in B, and set $\ell = |Q|$. Then every neighbour of x lies in B, and x lies in no triangle. If $\ell = 2$, then $d(x) = 1$. If $\ell \geq 3$, then $d(x) \leq \lfloor (\ell-1)/2 \rfloor$. In every case $\ell \leq q$, and $L(x) \leq c_q$ whenever $q \geq 5$.*

Proof. A neighbour of x in a component of $G - x$ that does not contain s would extend P, so x is not a cut vertex. Every edge with one end in B and one end outside B is incident with a cut vertex of B. Therefore every neighbour of x lies in B. The block is bipartite, so x lies in no triangle.

Since the block-cut tree is a tree, P enters B only once. The intersection of P with B is a single subpath $Q = z_0, \ldots, z_{\ell-1}$ with $z_{\ell-1} = x$. If s lies in B, then $Q = P$. If s lies outside B, the vertices of P that precede z_0 lie outside B, and z_0 is their only neighbour in B: any other edge from that prefix into B would be a second entry. A vertex of $B \setminus \{z_0\}$ is therefore adjacent to no vertex of the prefix. An extension of P at x through a vertex of B can fail only by a chord into Q.

Let y be a neighbour of x other than $z_{\ell-2}$. Then y lies in B, and y does not lie on Q, because Q is induced. The vertex y is not adjacent to the prefix. If y had no neighbour on Q, the path P followed by y would be induced. Thus y has a neighbour z_i on Q. This neighbour is not x. It is not $z_{\ell-2}$, for $yxz_{\ell-2}$ would be a triangle, and it is not $z_{\ell-3}$, for $z_{\ell-3}z_{\ell-2}xy$ would be a 4-cycle. Since B is bipartite, z_i has the colour of x, so $i \equiv \ell - 1 \pmod 2$ and $i \leq \ell - 5$.

If $\ell = 2$, then Q is a single edge and there is no admissible index, so $d(x) = 1$. If $\ell \geq 3$, the number of admissible indices is $\lfloor (\ell-1)/2 \rfloor - 1$. Two neighbours of x are nonadjacent and cannot share a blocker, for a common blocker would form a 4-cycle with them and with x. Hence $d(x) \leq \lfloor (\ell-1)/2 \rfloor$. In every case $\ell \leq q$. For $q \geq 5$ one has $d(x) \leq \lfloor (q-1)/2 \rfloor$, and $L(x) = 2d(x) \leq c_q$.

The same count applies inside B. Let z be the cut vertex of B lying on every path from a fixed root to B, or the root itself if the root lies in B, and let Q be a longest induced path of B starting at z, with other end x and with $\ell = |Q|$. A neighbour of x in B that had no neighbour on Q would extend Q inside B. If the root lies outside B, no vertex of $B \setminus \{z\}$ meets the component of the root at z, so that component contributes no further blocker. Thus $d_B(x) \leq 1$ if $\ell = 2$, and $d_B(x) \leq \lfloor (\ell-1)/2 \rfloor$ if $\ell \geq 3$. □

Theorem 8.2. (1) *Every terminal instance in $\mathcal{B}$ satisfies $S(G) \leq 2(D + q - 2) - m + c_q R$.*

(2) *Every $F \in \mathcal{B}$ with a vertex s such that $p_s(F) \leq q$ satisfies*

$$S(F) \leq c_q(n-1) - (c_q - 2)D - m.$$

In particular ROOT(q) *holds in* $\mathcal{B}$ *for every* $q \geq 3$. *If every block is bipartite or a triangle, one may take* $c_q = 2\lfloor (q-1)/2 \rfloor$ *for all* $q \geq 3$.

Proof. Two terminal facts. An endpoint x of a longest rooted induced path is not a cut vertex: a neighbour of x in another component of $G-x$ would extend the path. If z is at distance $q-1$ from s and $v_0 = s, \ldots, v_{q-1} = z$ is a geodesic, any neighbour $y \neq v_{q-2}$ of z must meet an earlier path vertex (else the rooted path extends); meeting v_i with $i \leq q-4$ shortens $d(s,z)$, and meeting v_{q-3} gives a C_4; so $y \sim v_{q-2}$, and two such y would give a C_4. Hence $L(z) = 2d(z) - 3t(z) \leq 2$. Moreover, if $x \neq s, z$, $G-x$ is connected and z survives, then $q-1 \leq d_{G-x}(s,z) \leq p_s(G-x) - 1 \leq q-1$, so the instance stays terminal with the same q.

Endpoint losses. Let x be the endpoint of a longest rooted path of order q. It lies in a single block. If that block is a triangle the loss $L(x)$ is 1, if an edge 2, and if a cycle of length at least 5 it is 4. If the block is bipartite, Lemma 8.1 gives $L(x) \leq c_q$ for $q \geq 5$.

No nontrivial bipartite block when $q \leq 4$. Let B be a nontrivial bipartite block, and let z be the cut vertex of B that lies on every path from s to B, or let $z = s$ if $s \in B$. Let Q be a longest induced path of B starting at z, and let x be its other end. Every vertex of a nontrivial block has degree at least two in the block, so $d_B(x) \geq 2$. The last paragraph of Lemma 8.1 gives $d_B(x) \leq 1$ if $|Q| \leq 4$ and $d_B(x) \leq \lfloor (|Q|-1)/2 \rfloor$ in any case with $|Q| \geq 3$. The inequality $d_B(x) \geq 2$ therefore forces $|Q| \geq 5$. An induced s–z path meets B only at z. Concatenating it with Q yields an induced path that starts at s. The order of this path is $|Q|$ if $s \in B$, and at least $|Q|+1$ if $s \notin B$. In either case $p_s(G) \geq 5$. A terminal instance with $q \leq 4$ therefore has no nontrivial bipartite block. Every block is an edge, a triangle, or a cycle of length at least five, so the graph is a cactus.

Bases. For $q \leq 4$ the preceding paragraph leaves only cacti. For a C_4-free cactus with c_3 triangle blocks and c_L longer cycle blocks, $S = 2(n-1) + 2c_L - c_3$ and $m \leq c_3$. For $q = 3$ no longer cycle occurs, and (1) follows. For $q = 4$ every longer cycle is a C_5 through s. A chosen length-3 geodesic uses at most one edge of each such C_5: if it used two edges and then left the cycle, replacing that two-edge arc by the other three-edge arc would give an induced rooted path of order 5. So each C_5 contributes two vertices to R, $R \geq 2c_L$, and $[2(D+2) - m + 3R] - S = R - 2c_L + c_3 - m \geq 0$. For $q = 2$ the root is universal and $S = 2D - m$.

Induction on n. Let E be the set of endpoints of rooted induced paths of order q and z a farthest vertex, so $z \in E$. If some $x \in E \setminus \{z\}$ exists, delete it: $x \notin N[s]$, x is not a cut vertex, its loss is at most c_q, the instance remains terminal with the same q, D and m, and R drops by one. If $E = \{z\}$, delete z: the new rooted maximum is $q-1$, the eccentricity $q-2$, R is unchanged and the loss is at most 2; induction gives $S \leq 2(D+q-3) - m + c_{q-1}R + 2 \leq 2(D+q-2) - m + c_qR$. Deletion preserves membership in $\mathcal{B}$. This proves (1); the bipartite-or-triangle variant is identical.

(2) Attach a new path with $q-1$ edges at s. The result is terminal of index q, lies in $\mathcal{B}$, has order $n+q-1$, mass $S + 2(q-1)$ and root degree $D+1$, and m is unchanged. Substituting into (1) gives (2). Since $c_q \leq q-1$ and $n-1-D \geq 0$,

$$\begin{aligned} c_q(n-1) - (c_q-2)D - m &= c_q(n-1-D) + 2D - m \\ &\leq (q-1)(n-1-D) + 2D - m \\ &= (q-1)(n-1) - (q-3)D - m, \end{aligned}$$

so ROOT(q) follows. □

A matching upper bound of shape $S \leq (r-1)n - 2r(r-3)$, for graphs in $\mathcal{B}$ of radius $r \geq 5$ and path order $2r-1$, has been proposed from a structural description of radius-extremal graphs. That description is not proved in this paper, and the radius bound in $\mathcal{B}$ is not claimed.

Lemma 8.3. *Let* G *be strong* C_4*-free and triangle-free, and let* $x_0x_1 \cdots x_k$ *be an induced path that cannot be extended at* x_k. *If* $d(x_k) \geq 2$, *then* $k \geq d(x_k) + 1$. *If moreover* G *is bipartite, then* $k \geq 2d(x_k)$.

Proof. Let $y \neq x_{k-1}$ be a neighbour of x_k. It is not on the path, and, since the path cannot be extended, y has a neighbour x_i with $i \leq k-1$. Here $i \neq k-1$ (a triangle) and $i \neq k-2$ (the 4-cycle $x_{k-2}x_{k-1}x_k y$), so $i \leq k-3$. A vertex x_i is adjacent to at most one neighbour of x_k. Hence the $d(x_k) - 1 \geq 1$ such neighbours need distinct blockers among $x_0, \ldots, x_{k-3}$, and $k-2 \geq d(x_k) - 1$. If G is bipartite, the blocker has the colour of x_k, so $i \equiv k \pmod 2$ and $i \leq k-4$; there are $\lfloor k/2 \rfloor - 1$ such positions, and $\lfloor k/2 \rfloor \geq d(x_k)$. □

Proposition 8.4. *Every connected regular bipartite strong C_4-free graph satisfies GEN_K for every K and satisfies REAL-133. Every Moore graph of diameter 2 and girth 5 satisfies REAL-133.*

Proof. Let G be d-regular. If $d = 1$ then $G = K_2$, with $r = 1$, $\ell = 1$, $p = 2$ and $S = n$, and both statements hold. Let $d \geq 2$. As G is triangle-free, $\lambda \equiv d$ and $\ell = d$. Lemma 8.3 applied to a longest induced path gives $p \geq 2d + 1$, and Theorem 2.4 gives $p \geq 2r - 1$; so $p \geq r + d$ both when $r \leq d + 1$ and when $r \geq d + 1$. If $p \leq 2K$, then $2d + 1 \leq 2K$, so $S = dn < Kn$. A Moore graph of degree $d \geq 2$ is triangle-free and strong C_4-free with $r = 2$ and $\ell = d$, and the first part of Lemma 8.3 gives $p \geq d + 2 = r + \ell$. □

Remark 8.5. A Moore graph of degree 57, if it exists, has $n = 57^2 + 1 = 3250$ and $S = 57n$. It satisfies REAL-133 by Proposition 8.4, which only needs $p \geq 59$. The statement GEN_{57} holds for it with equality in the mass bound, but GEN_{56} asks for $p \geq 113$ because $S > 56n$. We do not know whether such a graph would have $p \leq 112$. If it did, it would refute GEN_{56} while satisfying REAL-133. Its existence would therefore not by itself separate the two statements, but it shows that a proof of every GEN_K is a different assertion from REAL-133.

9. Negative results

The examples in this section are explicit graphs.

Example 9.1. Let $s \geq 3$, $t \geq 3$ and $m \geq 1$ be integers. Let $G_{s,t,m}$ be the cycle $C_{2s} = c_0 \cdots c_{2s-1}$ with a pendant path of t edges at every cycle vertex, together with m internally disjoint paths $c_0 x z y c_1$. Then $n = 2s(t+1) + 3m$, $e = 2s(t+1) + 4m$ and $T = 0$, so $e - n - T = m \geq 1$. For $(s, t, m) = (3, 3, 1)$ one has $n = 27$, $e = 28$, $r = 6$ and $p = 11$. These graphs satisfy REAL-133, since $\ell = 2e/n < 8/3$. They also show that $e = n + T$ fails for radius-extremal strong C_4-free graphs of every radius $r \geq 6$: here $r(G_{s,t,m}) = s + t$ and $p(G_{s,t,m}) = 2(s + t) - 1$.

Two vertices have at most one common neighbour, so G is strong C_4-free and triangle-free. An added path of length four does not shorten distances among the older vertices, and c_0 remains within distance two of every vertex of such a path, so the eccentricity of c_0 is still $s + t$. A vertex w of an added path meets the tip of the arm at c_s only through c_0 or c_1, hence at distance at least $s + t$. Thus $r = s + t$. For the path order, let B be the block formed by C and the added paths. An induced path P meets B in a subpath Q and can leave B only at an end of Q that is a cycle vertex, through at most t arm vertices. The same case division as for a single ear shows $|P| \leq 2s + 2t - 1$, using $t \geq 3$ and $s \geq 3$, and the cycle minus one vertex together with the arms at its two ends attains this order.

The last sentence of Example 9.1 abbreviates a case division which we now record. If Q avoids both c_0 and c_1, then $|Q| \leq 2s-2$. If Q contains exactly one of them, then either Q lies on the cycle with $|Q| \leq 2s - 1$, or $|Q| \leq 2s + 2$ with at most one cycle end, or $|Q| \leq 7$ with no cycle end. If Q contains both, the same three shapes occur with $|Q| \leq 2s-1$, $|Q| \leq 2s+2$ and $|Q| \leq 6$. Adding at most t arm vertices at each cycle end of Q gives $|P| \leq \max\{2s-1+2t,\ 2s+2+t,\ 7\} = 2s+2t-1$.

Example 9.2. Forbidding induced paths on eleven vertices and requiring $L(v) \geq 6$ at every vertex does not forbid an induced star $K_{1,6}$. Two Petersen graphs sharing a vertex give a triangle-free strong C_4-free graph on $n = 19$ vertices with $S = 60$, minimum deletion loss 6, $p = 9$ and $\lambda = 6$

at the shared vertex. For $b \geq 2$, gluing b copies of the Petersen graph along a common edge gives a 2-connected graph B_b with $n = 8b + 2$, $S = 28b + 2$, minimum loss 6, $p = 9$ and $\lambda = 2b + 1$ at the two hubs. Thus singleton losses do not bound λ. These graphs are not counterexamples to GEN_5: the eight non-hub vertices of one Petersen block have deletion loss $28 < 5 \cdot 8 + 1$.

For an oriented edge (u, v) of a connected strong C_4-free graph G, let q be the maximum order of an induced path whose first two vertices are u, v in that order, and let E be the set of terminal vertices of the induced paths of that order. The endpoint inequality $\mathrm{EP}(q)$ asserts $D_G(E) \leq q|E|$ for every such oriented edge of maximum order q. For a root s with $p_s(G) = q$, if E is the set of terminal vertices of the induced paths of order q starting at s, the rooted inequality $\mathrm{EP}^+(q)$ asserts $D_G(E) \leq (q-1)|E|$.

Proposition 9.3. *For every integer $q \geq 4$, $\mathrm{EP}(q)$ holds for all connected strong C_4-free graphs if and only if $\mathrm{EP}^+(q-1)$ does.*

Proof. Both graphs below remain strong C_4-free, so $S = 2e - 3T$ and the deletion loss of a vertex set is determined by the edges and triangles that meet it.

First assume $\mathrm{EP}^+(q-1)$, and let (u, v) be an oriented edge of maximum induced-path order q, with terminal set E. Delete $W = N[u] \setminus \{v\}$, and let H be the component of v in $G - W$. An induced path of G that starts with uv has no further vertex in $N[u]$, because the path is induced, so the remainder is an induced path of H starting at v. Conversely, every induced path of H starting at v extends by u to an induced path of G, since no vertex of H other than v is adjacent to u. Thus $p_v(H) = q - 1$, and the terminal set in H is still E.

Each $x \in E$ has at most one neighbour in W: any such neighbour is adjacent to u, and two of them would be two common neighbours of u and x. Every neighbour of x outside H lies in W. Let m_W be the number of edges between E and W, so $m_W \leq |E|$, and let T_{extra} be the number of triangles that meet E and are not contained in H. Every such triangle uses one of those edges. Comparing edges and triangles,

$$D_G(E) = D_H(E) + 2m_W - 3T_{\mathrm{extra}} \leq D_H(E) + 2|E|.$$

The rooted inequality in H is $D_H(E) \leq (q-2)|E|$, because $\mathrm{EP}^+(q-1)$ contributes the factor $(q-1)-1$. Therefore $D_G(E) \leq q|E|$.

Conversely, assume $\mathrm{EP}(q)$, and let (H, v) be a connected strong C_4-free graph with $p_v(H) = q-1$ and terminal set E. Form G by adding a vertex u adjacent to v and, for each $x \in E$, a new vertex a_x adjacent exactly to u and to x. Every terminal x fails to be adjacent to v: an induced path of order $q - 1 \geq 3$ from v to x has no chord vx. The vertex a_x has degree two. Any cycle through a_x uses both edges ua_x and $a_x x$, and then continues from x to the other neighbour of u on the cycle. That other neighbour is v or some a_y. In the first subcase the x–v path in H has length at least two, so the cycle has length at least five. In the second the cycle has the form $u, a_x, x, \ldots, y, a_y$ and again has length at least five. A cycle through u that avoids every a_x would need two neighbours of u in H, but the only neighbour of u in H is v. Thus G is strong C_4-free. No new triangle appears: the two neighbours of a_x are nonadjacent, and the neighbours of u are pairwise nonadjacent.

An induced path that starts with uv contains no a_x. Otherwise the edge ua_x would be a chord, since a_x is not the successor of u on that path. The induced uv-paths are therefore the paths u followed by an induced path of H starting at v. Their maximum order is q, and their terminal set is E. Exactly $|E|$ new edges meet E, namely the edges $a_x x$, and no new triangle meets E. Hence $D_G(E) = D_H(E) + 2|E|$. The oriented inequality gives $D_G(E) \leq q|E|$, so $D_H(E) \leq (q-2)|E|$, which is $\mathrm{EP}^+(q-1)$. □

Example 9.4. EP(9) fails. Let G have vertices $0, \dots, 14$ and adjacency lists

0: $4, 8, 11$	1: $5, 8, 10$	2: $6, 8, 9$	3: $7, 10$
4: $0, 9, 10$	5: $1, 9, 11$	6: $2, 10, 11$	7: $3, 11$
8: $0, 1, 2, 12$	9: $2, 4, 5$	10: $1, 3, 4, 6, 13$	11: $0, 5, 6, 7, 14$
12: $8, 13, 14$	13: $10, 12$	14: $11, 12$.	

The graph is triangle-free and strong C_4-free, with $n = 15$, $e = 23$ and $S = 46$. For the oriented edge $(12, 8)$ the maximum order is $q = 9$, attained only by

$$12, 8, 2, 9, 4, 10, 3, 7, 11 \qquad \text{and} \qquad 12, 8, 2, 9, 5, 11, 7, 3, 10.$$

Thus $E = \{10, 11\}$ and $D_G(E) = 20 > 18$. The same computation gives a failure of $\mathrm{EP}^+(8)$ on the subgraph induced by $0, \dots, 11$, rooted at 8: $p_8 = 8$, $E = \{10, 11\}$ and $D(E) = 16 > 14$. The inequalities EP(8) and $\mathrm{EP}^+(6)$ are not settled here.

Example 9.5. Let $k = p - r$. Call an induced subgraph H of G an *eligible core* if it is proper, connected and dominating, and $p(H) \le p(G) - 2$. The budget CB asks for an eligible core with

$$S(G) - S(H) \le k\big(n - |H|\big) + |H|.$$

Together with REAL-133 for H and an induction on the order, the budget would imply REAL-133. The Petersen graph has $S = 30$, $p = 5$, $r = 2$ and $k = 3$, hence $\ell = 3$. An eligible core has $p(H) \le 3$. A connected triangle-free strong C_4-free P_4-free graph is a star, by Lemma 2.7 and the absence of triangles. In the Petersen graph a star with $k \le 3$ leaves has centre of degree 3. The vertices it dominates are the centre, its three neighbours, and at most two further neighbours of each leaf, at most $4 + 2k$ vertices in all. This is 10 only for $k = 3$, so the eligible cores are exactly the ten closed neighbourhoods $N[v]$. Each has $S(H) = 6$, and $S(G) - S(H) = 24 > 3 \cdot 6 + 4 = 22$.

The unrestricted budget therefore fails. The example has $\ell = 3$. It does not determine a cutoff such as $\ell > 5$, and this paper contains no census of eligible cores.

Example 9.6. The degree, triangle and distance data of the Heawood graph ($n = 14$, cubic, $T = 0$, $r = 3$, $S = 42$, and $Z = 28$ nonadjacent pairs with no common neighbour) satisfy the counting identities used above, including $\sum_v d(v)^2 = n(n-1) + 6T - 2Z$, together with the lower bounds $p \ge 2r - 1$ and $p \ge \min_v \lambda(v) + 2$, if p is replaced by the value 5. But $5 < r + \ell = 6$. This system of scalar constraints does not imply REAL-133. The Heawood graph itself has $p = 7$.

Example 9.7. Let F be the incidence graph of the projective plane $\mathrm{PG}(2, q)$, with $f = |F| = 2(q^2 + q + 1)$ and $S(F) = (q+1)f$, and choose a root of F. Let $t = f - 1$ and $b \ge 1$. Take a 6-cycle $c_0 \cdots c_5$, attach a pendant path with t edges at each cycle vertex, and attach b copies of F at c_0 by identifying their roots with c_0. The result is strong C_4-free, because each new block meets the rest in a cut vertex, and

$$n = 6f + b(f - 1), \qquad S = 12f + b(q + 1)f, \qquad r = t + 3, \qquad p = 2t + 5 = 2r - 1.$$

Distances inside the flower are unchanged. Every vertex of an attached copy is within distance $f - 1 \le t$ of c_0, so each cycle vertex has eccentricity $t + 3$, while a vertex x of a copy has distance $d_F(x, c_0) + t + 3$ from the tip of the arm at c_3. An induced path meets the cycle in at most five consecutive vertices, and each of its two ends outside the cycle contributes at most t vertices. Mass is additive at cut vertices by Lemma 2.3. For fixed q, as $b \to \infty$ one has $\ell \to (q + 1)f/(f - 1)$. Letting $q \to \infty$, with $r = f + 2$, this limit grows like $\sqrt{r/2}$. For $q = 5$ and $b = 17$ one has $n = 1409$, $r = 64$, $p = 127$ and $S = 7068 > 5n$. These graphs satisfy $\mathrm{RAD}(r)$ for their own radius, and they show that ℓ is unbounded on the class $p = 2r - 1$. Any theorem $\mathrm{RAD}(r)$ must use a bound that depends on r.

10. What is proved and what is open

The comparisons with Chen [3], Chen and Xu [5], Chen, Wu and Xu [6], Chudnovsky and Stacho [4], DeLaViña, Larson, Pepper and Waller [8], Karthick and Maffray [12], and Brandstädt and Hoàng [2] are in the introduction, in Section 5 and in Remarks 3.7 and 3.8. Table 1 is the only status list. An open row is not a hypothesis of a proved row. A finite search is not a proof here.

For $5 < \ell \leq 6$, Theorem 6.1 gives $2r - 1 \leq p \leq r + 5$ and $r \leq 6$. Lemma 3.1 excludes radius 1, Theorem 3.6 excludes $p \leq 5$, and Corollary 4.4 excludes $(4, 7)$. Lemma 5.6 excludes $(2, 6)$ and $(3, 6)$. For bipartite graphs, Lemma 6.3 excludes $(2, 7)$ and Lemma 6.4 excludes $(3, 7)$. The pairs that remain are necessary conditions for a counterexample in this range. The table does not assert that any of them occurs.

Statement	Status
$M(n)$ for connected strong C_4-free P_6-free graphs	proved, Theorem 3.6
the structure of graphs with $r = 4$ and $p = 7$, the formula $S = 2n - b - m$, and the equality family	proved, Theorem 4.3 and Corollaries 4.4, 4.5
$2e \leq 3n$ for connected triangle-free strong C_4-free P_7-free graphs, equality only for the Petersen graph	proved, Theorem 5.4
$e \leq 2n - 3$ for connected strong C_4-free P_7-free graphs, hence $\ell < 4$, and $\ell \leq 4 - 9/n$ if a triangle is present	proved, Lemma 5.6
REAL-133 for $\ell \leq 3$; for $r \leq 2$ and $\ell \leq 4$; for $r = 4$ and $\ell \leq 4$; for $r \geq 5$ and $\ell \leq 4$	proved, Theorem 5.9
ROOT(3) and ROOT(4); the terminal strengthening of Proposition 7.3	proved
ROOT(q) for every q, in the class $\mathcal{B}$	proved, Theorem 8.2
a minimum counterexample to GEN_4 is 2-connected	proved, Proposition 7.4; GEN_4 is open
the cell $r = 3$, $p = 6$, $3 < \ell \leq 4 - 9/n$	open; it contains a triangle, and $\ell = 4$ does not occur, Corollary 5.7
GEN_3 for graphs that contain a triangle	open; Appendix A is conditional
GEN_K for $K \geq 4$, unrestricted GEN_6, RAD(r) for $r \geq 5$, and ROOT(q) for $q \geq 5$ outside $\mathcal{B}$	open
REAL-133	open
pairs (r, p) still necessary for $5 < \ell \leq 6$	$(2, 7)$, $(3, 7)$, $(3, 8)$, $(4, 8)$, $(4, 9)$, $(5, 9)$, $(5, 10)$, $(6, 11)$
pairs excluded from $5 < \ell \leq 6$ for every strong C_4-free graph	$(2, 6)$ and $(3, 6)$, by Lemma 5.6
further pairs excluded from $5 < \ell \leq 6$ for bipartite graphs	$(2, 7)$ by Lemma 6.3; $(3, 7)$ by Lemma 6.4

Table 1. Proved statements and statements left open. The last three rows are the corrected account of the range $5 < \ell \leq 6$.

No structure theorem for connected strong C_4-free graphs of radius $r \geq 5$ and path order $2r - 1$ is proved here. Notes on deletion in the triangle-free class, and on ports of a distance-preserving dominating set, are kept outside this article in `notes/supplement-deletion-ports.tex`.

Conjecture 10.1. ROOT(q) *holds for every* $q \geq 3$. *For every* $r \geq 5$, *every connected strong* C_4*-free graph of radius* r *and path order* $2r - 1$ *satisfies* $S \leq (r - 1)n - 2r(r - 3)$, *with equality only for* C_{2r}.

The cases $q = 3$ and $q = 4$ are Lemma 5.3 and Theorem 7.2, and every q holds in $\mathcal{B}$ by Theorem 8.2. The radius bound is not claimed for $r \geq 5$.

Conjecture 10.2. *If F is connected and strong C_4-free, $p_s(F) = q$ and the eccentricity of s is $q-1$, then*

$$S(F) \leq (q-1)(n-1) - (q-3)D - m - (q-2)(q-3).$$

For $q = 4$ the subtracted term is 2, and Proposition 7.3 is this case. For $q \geq 5$ the conjecture is open. The terminal bound in $\mathcal{B}$ is the first part of Theorem 8.2.

Conjecture 10.2, for a fixed q, implies ROOT(q). The implication does not prove the conjecture. Let F be connected and strong C_4-free with $p_s(F) \leq q$. Form G by attaching a path of $q-1$ new edges at s. Then G is terminal of index q: the new path is an induced path of order q starting at s, and every induced path of F that starts at s has order at most q. The order, the root degree and the mass increase by $q-1$, 1 and $2(q-1)$ respectively, and m is unchanged. Write $N = n+q-1$ and $D_G = D+1$. The terminal inequality in G reads

$$S(F) + 2(q-1) \leq (q-1)(N-1) - (q-3)D_G - m - (q-2)(q-3).$$

The identity $N - 1 = n + q - 2$ gives $(q-1)(N-1) = (q-1)(n-1) + (q-1)^2$, and $(q-3)D_G = (q-3)D + (q-3)$. The terms that do not involve n, D or m are

$$(q-1)^2 - (q-3) - (q-2)(q-3) - 2(q-1) = (q-3)\big((q-1) - (q-1)\big) = 0.$$

What remains is $S(F) \leq (q-1)(n-1) - (q-3)D - m$, which is ROOT($q$). For $q = 4$ this route uses Proposition 7.3. The induction in Theorem 7.2 does not. For $q \geq 5$ the hypothesis is open, so the implication leaves ROOT(q) open.

Conjecture 10.3. *Every connected strong C_4-free graph G that is not a cycle and has $\ell(G) > 5$ has an eligible core H, in the sense of Example 9.5, with*

$$S(G) - S(H) \leq (p-r)\big(n - |H|\big) + |H|.$$

Example 9.5 shows that the budget without a restriction on ℓ is false, and that the Petersen graph is a counterexample with $\ell = 3$. The example does not show that 5 is the correct cutoff, and it does not count eligible cores. A reported count of cores of order at most 12 is not a result of this paper. If the conjecture, the range $\ell \leq 5$, and an induction on the order were all available, REAL-133 would follow. The range $\ell \leq 5$ is not a theorem, so this route is not a proof of REAL-133. Example 9.6 shows that a proof cannot rely only on the scalar constraints listed there.

Appendix A. A conditional reduction for GEN_3

Theorem 5.4 is the triangle-free case of GEN_3. The three local statements named at the end of this appendix are not proved for graphs that contain triangles. With them, the hand arguments below would give GEN_3 in general. Without them, that case remains open. The cell $r = 3$, $p = 6$, $3 < \ell \leq 4 - 9/n$ remains open as well. By Corollary 5.7 every graph in the cell contains a triangle, and $\ell = 4$ does not occur.

Theorem A.1. *Let G be a counterexample to GEN_3 of minimum order: G is connected and strong C_4-free, $p(G) \leq 6$, $S(G) \geq 3n+1$, and n is minimum. Then G is 2-connected; $S(G) - S(G-U) \geq 3|U| + 1$ for every nonempty proper $U \subset V(G)$; $2\lambda(v) - t(v) \geq 4$ for every v, so $\lambda(v) \geq 2$; and every vertex with $\lambda(v) = 2$ has degree 2 and nonadjacent neighbours, and no two vertices of degree 2 are adjacent.*

Proof. Since $S \leq 2 < 3n$ for $n \leq 2$, one has $n \geq 3$. Let v be a cut vertex. For each component C of $G - v$ let ρ_C be the largest order of an induced path in $G[C \cup \{v\}]$ with end v. Two such paths from different components form an induced path on $\rho_C + \rho_{C'} - 1 \leq 6$ vertices, so some component C_1 has

$\rho_{C_1} \le 3$. Put $G_1 = G[C_1 \cup \{v\}]$ and $G_2 = G - C_1$, with $n_i = |G_i|$, so $n = n_1 + n_2 - 1$ and $n_1 \ge 2$. Both graphs are connected, smaller than G, and P_7-free. Lemma 5.3 gives $S(G_1) \le 2(n_1 - 1)$, and minimality gives $S(G_2) \le 3n_2$. By Lemma 2.3, $S(G) \le 2(n_1 - 1) + 3n_2 = 3n - (n_1 - 1) < 3n$.

Every component H of $G - U$, for U nonempty and proper, is connected, smaller, and P_7-free, so $S(H) \le 3|H|$. Summing, $S(G - U) \le 3(n - |U|)$, and $S(G) \ge 3n + 1$ gives the deletion bound. The case $U = \{v\}$ and Lemma 2.2 give $2\lambda(v) - t(v) \ge 4$. If $\lambda(v) = 2$, then $t(v) = 0$, so $d(v) = 2$ and the neighbours are nonadjacent. Every degree-2 vertex has $\lambda = 2$ and hence lies in no triangle. If $u \sim w$ both have degree 2, with other neighbours x and y, then $x \ne y$ because $t(u) = 0$. The set $\{u\}$ is a unit of x and $\{w\}$ is a unit of y, so Lemma 2.2 with $U = \{u, w\}$ gives loss $2 + 2 + 1 + 1 = 6 < 7$. □

The same mass bound for a disconnected graph is the sum of the bounds on its components. An isolated vertex has $S = 0$. Call a graph *admissible* if it is strong C_4-free and P_7-free, $L(v) \ge 4$ for every vertex, and no two adjacent vertices have degree 2. A minimum counterexample is admissible. In an admissible graph, $\lambda(v) = 2$ forces $t(v) = 0$ and $d(v) = 2$. Write B for the vertices of degree at least 3, the *branch vertices*, and Z for the degree-2 vertices, the *ears*. Put $D = G[B]$, $d_D(v) = |N(v) \cap B|$ and $q(v) = |N(v) \cap Z|$. All triangles lie in D.

Lemma A.2. *In an admissible graph, $q(v) \le 4$ for every vertex v.*

Proof. Let u_i be ears at v with other neighbours a_i. The a_i are distinct, nonadjacent to v, and of degree at least 3. Put $E_i = N(a_i) \setminus \{u_i\}$. Then $|E_i| \ge 2$ and $E_i \cap N[v] = \emptyset$.

If $a_i \not\sim a_j$, then $E_i = \{w, p\}$ and $E_j = \{w, q\}$ with $pq \in E$, $w \not\sim p$ and $w \not\sim q$. Indeed two common neighbours of a_i and a_j are forbidden, so $|E_i \cap E_j| \le 1$. Since $|E_i| \ge 2$, the private part $E_i \setminus E_j$ is nonempty, and likewise $E_j \setminus E_i$. If p lies in the first private part and q in the second with $p \not\sim q$, the path $pa_iu_ivu_ja_jq$ is induced. Thus the private parts are complete to each other. If one of them contained two vertices, those two vertices and one vertex of the other private part would form a 4-cycle with the corresponding a-vertex. Both private parts therefore have size one, and $|E_i \cap E_j| = 1$. An edge from the common vertex to a private vertex is another 4-cycle. In particular a_i and a_j have degree 3 and lie in no triangle.

No three of the vertices a_i are pairwise nonadjacent. Otherwise the sets E_i meet pairwise in a single vertex. If the three intersections are the three pairs of a triple, the claim forces a triangle on that triple, and one a_i together with two vertices of the triple forms a 4-cycle. Otherwise $E_i = \{w, p_i\}$ with $p_1p_2p_3$ a triangle and w nonadjacent to each p_i, and $vu_1a_1wa_2p_2p_3$ is an induced P_7.

Five ears are impossible. The graph F they determine on $\{a_1, \dots, a_5\}$ has independence number at most two. If some vertex a of F is adjacent to the other four, those four induce a matching. The independence number forces that matching to be perfect, say $\{b_1b_2, c_1c_2\}$. Then $b_1 \not\sim c_1$, so the preceding paragraph says that b_1 lies in no triangle, but ab_1b_2 is a triangle. Otherwise every vertex of F has a non-neighbour, hence has degree 3 in G and lies in no triangle. One neighbour of each such vertex is its ear, so F has maximum degree at most two. It is triangle-free and has no 4-cycle, and its independence number is at most two, so F is a 5-cycle. Label that cycle $a_1a_2a_3a_4a_5a_1$, and write u_i for the ear joining v to a_i. The seven vertices $u_5, v, u_1, a_1, a_2, a_3, a_4$, in that order, induce a P_7: each u_i has neighbours v and a_i only, v is nonadjacent to every a_j, and the only edges of the cycle among a_1, a_2, a_3, a_4 are the three path edges. □

Proposition A.3. *Let G be admissible, with B, Z, D, d_D and q as above. Suppose that $d_D(v) \le 3$ for every v, that $d_D(v) = 3$ implies $q(v) = 0$, and that $d_D(v) = 2$ with the two neighbours in B nonadjacent implies $q(v) \le 2$. Then $S(G) \le 3n$.*

Proof. Ears have nonadjacent neighbours and Z is an independent set, so $S(G) = S(D) + 4|Z|$. Since $|Z| = \frac{1}{2}\sum_{v\in B} q(v)$ and $n = |B| + |Z|$,

$$S - 3n = \sum_{v\in B}\Big(d_D(v) - t_D(v) - 3 + \tfrac{1}{2}q(v)\Big).$$

Lemma A.2 gives $q(v) \le 4$. If $d_D(v) = 2$ and $t_D(v) = 0$, the two neighbours in B are nonadjacent, for an edge between them would be a triangle. The three suppositions and the ear bound make every summand nonpositive: the summand is at most -1 when $(d_D, t_D) = (0, 0)$, at most 0 when $(d_D, t_D) = (1, 0)$, at most 0 when $(d_D, t_D) = (2, 0)$, at most 0 when $(d_D, t_D) = (2, 1)$, and equal to $-t_D(v) \le 0$ when $d_D(v) = 3$. The suppositions exclude $d_D(v) \ge 4$. □

The three suppositions in Proposition A.3 are not proved in this paper. Theorem A.1 shows that a minimum counterexample to GEN_3 is admissible, so those suppositions would imply GEN_3. They do not, by themselves, make GEN_3 a theorem.

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